A Bipartite Graph Partition-Based Coclustering Approach With Graph Nonnegative Matrix Factorization for Large Hyperspectral Images

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***Abstract*— Clustering large hyperspectral images (HSIs) is a very challenging problem because large HSIs have high dimensionality, large spectral variability, and large computational and memory consumption. Recently, sparse subspace clustering (SSC) has achieved remarkable success in HSI clustering. However, most SSC-based methods suffer from the following bottlenecks for large HSIs: 1) high computational consumption and memory space during the construction of the similarity matrix and decomposition of the graph Laplacian matrix and 2) failure to capture the relationships among dictionary atoms, sparse coefficients, and hyperspectral pixels. To address these challenges, we propose a novel algorithm that extends SSC to cocluster large HSIs, called bipartite graph partition with graph nonnegative matrix factorization (BGP-GNMF). Specifically, to fully explore the characteristics of the spectral and spatial contexts in HSIs, we propose a novel superpixel and pixel coclustering framework with bipartite graph partitioning in the joint sparse representation domain, where superpixel-based dictionary atoms are defined as disjoint vertex sets of the bipartite graph and the joint sparsity representation is mapped into the adjacency matrix of the undirected bipartite graph. To overcome the challenges of high computational consumption and large memory space for large HSIs, the bipartite graph partition with orthonormal constrained nonnegative matrix factorization is proposed to simultaneously cluster the structured dictionary atoms and hyperspectral pixels with an indicator matrix. Finally, to exploit the intrinsic geometry of HSIs, we incorporate manifold regularization into the bipartite graph partition to improve final clustering accuracy. The effectiveness and efficiency of the proposed method are verified on three classical HSIs, and the experimental results illustrate the superiority of the proposed method compared with other state-of-the-art HSI clustering methods.**

***Index Terms*— Bipartite graph partition, dictionary learning, joint sparsity representation, nonnegative matrix factorization (NMF), sparse subspace clustering (SSC).**

I. Introduction

HYPERSPECTRAL remote sensors obtain hundreds of narrow contiguous spectral bands, which can be used to identify different materials more accurately than traditional panchromatic and multispectral remote sensing images [1]. In the application of hyperspectral remote images, unsupervised classification or clustering is a fundamental technique that has been widely applied in many fields, such as environmental monitoring [2], forest inventory [3], and land-use mapping [4]. Clustering in hyperspectral images (HSIs) assigns unique land-cover classes to each hyperspectral pixel by utilizing meaningful information. To date, many clustering methods have achieved great success, and according to the theories and working mechanisms, they can be roughly split into the following five clustering methods: centroid-based clustering [5], density-based clustering [6], distribution-based clustering [7], hierarchical-based clustering [8], and subspace-based clustering [9].

Specifically, centroid-based clustering methods iteratively update the centroids and assign the clusters for hyperspectral pixels until the partition error between the centroids and pixels meets the tolerance condition. In centroid-based clustering methods, fuzzy c-means clustering [10], k-means clustering [11], and k-medoids clustering [12] are typical and effective algorithms in practical applications. Density-based clustering methods locate regions of higher density to construct clusters, and the densities of the regions are calculated by the number of pixels within some radius. Density-based spatial clustering of applications with noise (DBSCAN) [6] and clustering by fast search and find of density peaks (CFSFDP) [13] are typical algorithms for density-based clustering methods. The methods of distribution-based clustering, such as the expectation–maximization clustering

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algorithm [14], suppose that pixels belonging to the same distribution are assigned into a single cluster. Distribution-based clustering methods can explore the properties of dependence and correlation between the attributes of pixels. The hierarchical-based clustering methods seek to build a clustering hierarchy by recursively segmenting the hyperspectral pixels in either a top-down or bottom-up method until the tolerance condition is met. The classical methods of hierarchical-based clustering include agglomerative clustering [8] and divisive hierarchical clustering [15]. However, these methods have difficulty in dealing with complex nonlinear structured datasets and irregularly shaped clusters.

Recently, subspace-based clustering methods have achieved a significant advantage in handing complex structured high-dimensional datasets. The goal of the subspace-based methods is to find multiple low-dimensional subspaces and the cluster membership of each data point. The subspace-based clustering methods create a low-dimensional embedding of the affinity matrix and then apply the centroid-based clustering method to the low-dimensional feature space [16]. The typical methods of subspace-based clustering are self-express subspace clustering and scalable subspace clustering. For self-express subspace clustering, the reconstruction coefficients of the self-representation model are used to construct an affinity matrix by imposing a prior or regularization on the coefficients. For example, low-rank subspace clustering (LRSC) [17], locality and structure regularized low-rank representation [18], sparse subspace clustering (SSC) [9], and l_2 norm regularized SSC [19] are typical methods. Scalable subspace clustering resolves the scalability issue for SSC. For instance, an out-of-sample extension of SSC was proposed to cluster large-scale data [20], a sketching method was proposed to speed up SSC [21], kernel SSC was proposed and incorporates spatial and spatial max pooling information for the nonlinear structure of HSIs [22], structured-enforced dictionary learning-based subspace clustering was proposed to reduce dimensional dictionaries and coefficient matrices [23], and low-rank and sparse-based robust subspace representations were presented for high-dimensional corrupted data [24]. An exemplar-based subspace clustering was proposed for imbalanced and large-scale datasets [25]. However, the subspace-based clustering methods mentioned above cluster samples according to their features or cluster features according to their distribution on the samples, and they ignore the duality and co-occurring cluster structure in the relationship between clusters of samples and clusters of features.

In recent years, popular coclustering methods have been well studied in the fields of text classification, document categorization, and gene expression. The goal of coclustering is to explore coherent patterns and co-occurrence by simultaneously clustering the samples and features of data. For instance, in the document collection coclustering framework, the problem is usually modeled as a bipartite graph between words and documents, in which the spectral graph partition is utilized to partition the bipartite graph model [26], [27]. Previous research shows that, coclustering outperforms one-way clustering, i.e., either sample or feature clustering. The underlying rationale is duality of sample and feature clustering; for example, sample clustering is able to promote the ability of feature clustering, while feature clustering can boost the performance of sample clustering. Under the coclustering framework, this significant advantage motivates us to consider HSI coclustering as a bipartite graph partition problem between pixels and a superpixel, in which the coclustering problem can be solved by spectral graph partitioning.

Although bipartite graph partition-based coclustering methods have achieved good performance, it is hard to obtain an optimal solution for spectral decomposition in processing the large affinity matrix of large HSIs [28]. In the bipartite graph partition, singular value decomposition (SVD) is required for the normalized affinity matrix, leading to great challenges in computational consumption for large and high-dimensional hyperspectral datasets. In addition, it is expensive to store an entire affinity matrix in large HSIs. Recently, nonnegative matrix factorization (NMF) has attracted increasing attention because it can find two or more lower dimensional matrices to approximate the original data well, solving the clustering problem of large and high-dimensional datasets. Especially in document clustering and face recognition, NMF has been proved to surpass SVD [29]. For example, Zhang *et al.* [30] proposed a robust low-rank matrix factorization to solve the high-dimensional features of HSIs in which clustering can be achieved by combining the dimensionality reduction and clustering into a unified framework. Cai *et al.* [31] constructed an affinity graph to capture the geometrical context of the original data, which is used to factorize a matrix with respect to the graph structure. Wang *et al.* [32] proposed a graph regularized NMF with bistochastic matrix to learn a high-quality graph, in which the graph regularized NMF and the iterative reweighted algorithm were incorporated into a robust loss function. Chris *et al.* [33] presented the equivalence of NMF and Laplacian-based spectral clustering.

Inspired by the aforementioned analysis, we propose a novel algorithm for coclustering large HSIs, called bipartite graph partition with graph nonnegative matrix factorization (BGP-GNMF) for large HSIs. The main contributions of this article are summarized as follows.

1) With superpixel-based dictionary learning and sparse representation, we propose a bipartite graph partition method to capture the intrinsic relationship between two disjoint sets: hyperspectral pixels and superpixel-based dictionary atoms. Due to the bipartite graph models, the complex correlations and inherent structure in HSIs can be fully exploited to strengthen the coclustering performance.
2) To address the challenging task of clustering large HSIs with high computational complexity and huge memory space, we propose a novel alternative algorithm for coclustering large HSIs, namely, BGP-GNMF. To exploit the intrinsic geometry of HSIs, a unified bipartite graph partition with manifold regularization model is proposed to promote the discriminating power of indicator matrix for the hyperspectral pixels.
3) Finally, an efficient multiplicative update rule (MUR) is adopted to design an optimization algorithm for BGP-GNMF. Extensive experimental results on three

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typical HSIs illustrate that the proposed coclustering model performs more effectively than the existing clustering methods for large HSIs.

The rest of this article is described as follows. Section II describes the related work of bipartite graph partitioning for HSIs in detail. Section III presents the coclustering for large HSIs. Section IV describes and analyzes the experimental results. Finally, Section V draws the conclusion of this article.

II. Related Works and Motivation

Aiming at assigning closely related entities to the same group, coclustering methods have received increasing attention in recent years. Compared with traditional SSC techniques, coclustering methods address the problem of simultaneous clustering of both dimensions of a data matrix by taking advantage of dual relationship information, such as the feature-sample matrix, term-document matrix, signal-dictionary matrix, and gene-sample matrix. In this article, we focus on hyperspectral pixels and dictionary atoms, which are modeled as two disjoint sets of the vertices of an undirected weighted bipartite graph.

Let $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_i, \ldots, \mathbf{y}_n] \in R^{d\times n}$ denote an HSI composed of n pixel vectors $\mathbf{y}_i$, where d is the number of spectral bands. Under the sparse representation framework, the hyperspectral pixels are expressed by a linear combination of learned dictionary atoms, and the typical dictionary learning problem can be represented in a sparsity-constrained matrix decomposition model

$$\arg\min_{\mathbf{B},\mathbf{X}} \frac{1}{2}\|\mathbf{Y}-\mathbf{B}\mathbf{X}\|_F^2 + \sum_{i=1}^{n}\lambda_i\|\mathbf{x}_i\|_p \tag{1}$$

where $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_i, \ldots, \mathbf{x}_n] \in R^{m\times n}$ denotes the sparse representation matrix corresponding to the dictionary $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_i, \ldots, \mathbf{b}_m] \in R^{d\times m}$, m denotes the number of dictionary atoms, and λ_i denotes the tradeoff parameter. The optimization is solved by updating the sparse coefficient and dictionary atoms until convergence.

1) *Updating the Sparse Coefficient* $\mathbf{X}$ *Fixed the Dictionary* $\mathbf{B}$*:* If $p = 0$, the solution is obtained by orthogonal matching pursuit (OMP) and matching pursuit (MP) [34], and if $p = 1$, the sparse coding optimization problem is also known as Lasso and uses the convex l_1 norm to approximate the l_0 norm.
2) *Updating Dictionary Atoms* $\mathbf{B}$ *With Given* $\mathbf{X}$*:* The dictionary atoms are achieved by minimizing the overall representation error, whose solution is solved by the K-singular value decomposition (K-SVD) method and method of optimal directions (MOD) [35].

From (1), it can be clearly observed that the sparse representation can encode the explicit association information between dictionary atoms and hyperspectral pixels. Under the sparse representation framework, we map two disjoint sets of vertices about hyperspectral pixels and dictionary atoms to an undirected weighted bipartite graph $G = (\mathcal{B}, \mathcal{Y}, E)$, where $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_i, \ldots, \mathbf{b}_m\}$ denotes the set composed of dictionary atoms, $\mathcal{Y} = \{\mathbf{y}_1, \ldots, \mathbf{y}_j, \ldots, \mathbf{y}_n\}$ denotes the set composed of hyperspectral pixels, and E denotes the set composed of edges $\{\{\mathbf{b}_i, \mathbf{y}_j\} : \mathbf{b}_i \in \mathcal{B}, \mathbf{y}_j \in \mathcal{Y}\}$, which encodes the pairwise similarities.

Let $\mathbf{A}$ be the nonnegative matrix of $\mathbf{X}$ with $\mathbf{A} = |\mathbf{X}|$. Then, we describe the adjacency matrix $\mathbf{M}$ of the weighted bipartite graph as follows:

$$\mathbf{M} = \begin{bmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^T & \mathbf{0} \end{bmatrix} \in R^{(m+n)\times(m+n)}. \tag{2}$$

The classical bipartite graph partition is used to find subsets of vertices that are almost equal size by performing graph cut methods, such as the ratio cut, the min-max cut, the min cut, and the normalized cut. As an example, we can apply the normalized cut on the graph $G = (\mathcal{B}, \mathcal{Y}, E)$ to achieve the goal of coclustering. Suppose that the dictionary atoms are partitioned into k subsets $\mathcal{B} = \{\mathcal{B}_1, \ldots, \mathcal{B}_k\}$, and the hyperspectral pixels are partitioned into k subsets $\mathcal{Y} = \{\mathcal{Y}_1, \ldots, \mathcal{Y}_k\}$; thus, the graph G is partitioned into k components $\mathcal{V} = \{\mathcal{V}_1, \ldots, \mathcal{V}_k\}$, where $\mathcal{V} = \mathcal{B} \cup \mathcal{Y}, \mathcal{V}_k = \mathcal{B}_k \cup \mathcal{Y}_k$. To this end, the objective function of the bipartite graph partition can be described as follows:

$$N_{\text{cut}} = \sum_{i=1}^{k} \frac{\text{cut}\left(\mathcal{V}_i, \bar{\mathcal{V}}_i\right)}{\text{weight}(\mathcal{V}_i)} \tag{3}$$

where $\bar{\mathcal{V}}_i$ is the complement of subset $\mathcal{V}_i$ in graph G, weight$(\mathcal{V}_i)$ denotes the accumulated edge weights within a group, $\text{cut}(\mathcal{V}_i, \bar{\mathcal{V}}_i) = \sum_{i\in\mathcal{V}_i}\sum_{j\in\bar{\mathcal{V}}_i}\mathbf{M}_{ij}$, and $\text{weight}(\mathcal{V}_i) = \sum_{i\in\mathcal{V}_i}\sum_{k}\mathbf{M}_{ik}$.

Assume that $\mathbf{Q} \in R^{(m+n)\times k}$ is the indicator matrix of bipartite graph G, where $\mathbf{q}_{ij} = 1$ denotes the ith node is partitioned into the jth component. The Rayleigh quotient of the indicator matrix $\mathbf{Q}$ is equal to the objective function value of (3), which is expressed as follows:

$$\min_{\mathbf{Q}} \sum_{i=1}^{k} \frac{\mathbf{q}_i^T\mathbf{L}\mathbf{q}_i}{\mathbf{q}_i^T\mathbf{D}\mathbf{q}_i} \tag{4}$$

where $\mathbf{q}_i$ is the ith column of matrix $\mathbf{Q}$, $\mathbf{L} = \mathbf{D} - \mathbf{M}$ denotes the Laplacian matrix corresponding to the adjacency matrix $\mathbf{M}$, and $\mathbf{D}$ denotes the diagonal matrix with $\mathbf{D} = \sum_j \mathbf{M}_{ij}$.

Assume that $\mathbf{F} = \mathbf{D}^{(1/2)}\mathbf{Q}(\mathbf{Q}^T\mathbf{D}\mathbf{Q})^{-(1/2)}$, $\mathbf{I}$ denotes the identity matrix, and (4) can be rewritten as

$$\min_{\mathbf{F}^T\mathbf{F}=\mathbf{I}} \mathbf{Tr}\left(\mathbf{F}^T\tilde{\mathbf{L}}\mathbf{F}\right) \tag{5}$$

where $\tilde{\mathbf{L}}$ denotes the normalized Laplacian matrix and $\tilde{\mathbf{L}} = \mathbf{I} - \mathbf{D}^{-(1/2)}\mathbf{M}\mathbf{D}^{-(1/2)}$. Again, if $\mathbf{F} = \begin{bmatrix}\mathbf{V}_1 \\ \mathbf{V}_2\end{bmatrix}$ and $\mathbf{D} = \begin{bmatrix}\mathbf{D}_1 & \\ & \mathbf{D}_2\end{bmatrix}$, we have

$$\max_{\mathbf{V}_1{}^T\mathbf{V}_1+\mathbf{V}_2{}^T\mathbf{V}_2=\mathbf{I}} \mathbf{Tr}\left(\mathbf{V}_1{}^T\mathbf{D}_1^{-\frac{1}{2}}\mathbf{A}\mathbf{D}_2^{-\frac{1}{2}}\mathbf{V}_2\right) \tag{6}$$

where $\mathbf{V}_1 \in R^{m\times k}$, $\mathbf{V}_2 \in R^{n\times k}$, $\mathbf{D}_1(i,i) = \sum_j \mathbf{A}_{ij}$, and $\mathbf{D}_2(j,j) = \sum_i \mathbf{A}_{ij}$. Note that the matrices $\mathbf{V}_1$ and $\mathbf{V}_2$ are simultaneously restricted to discrete values, which makes the problem in (6) difficult to solve. To solve this problem, we can relax the restriction on matrices $\mathbf{V}_1$ and $\mathbf{V}_2$ from discrete values to continuous ones, and the orthonormal constraint $\mathbf{V}_1{}^T\mathbf{V}_1 + \mathbf{V}_2{}^T\mathbf{V}_2 = \mathbf{I}$ still holds. In this way, Nie *et al.* [36] proved that the optimal solutions for $\mathbf{V}_1$ and $\mathbf{V}_2$ should be the leading k left and right singular vectors of $\mathbf{D}_1^{-(1/2)}\mathbf{A}\mathbf{D}_2^{-(1/2)}$.

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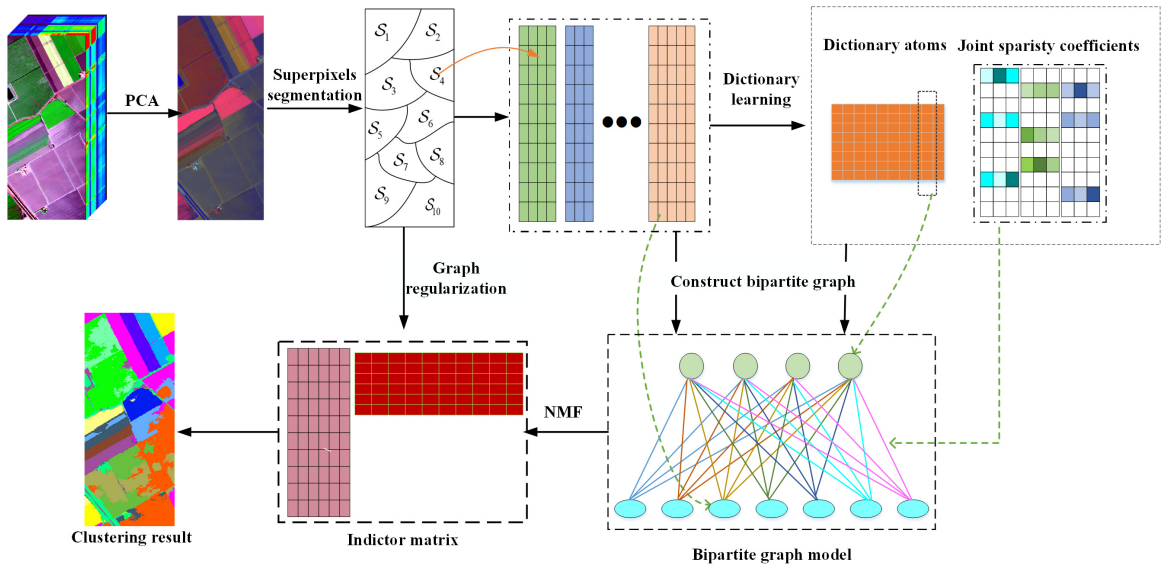


Fig. 1. Framework of the proposed coclustering method.

Next, the bipartite graph partitions can be obtained by running the k-means on the rows of $\mathbf{F}$. Finally, the bipartite graph partition algorithm is described in Algorithm 1.

Algorithm 1 BGP-DL for HSIs

Input: $\mathbf{Y} \in R^{d\times n}$: hyperspectral image, k: the number of clusters.

1: Perform dictionary learning (Eq. 1) to obtain the dictionary atoms $\mathbf{B}$

2: Construct a weighted bipartite graph $G = (\mathcal{B}, \mathcal{Y}, E)$

- Define disjoint vertices sets $\mathcal{B}$ and $\mathcal{Y}$ by dictionary atoms $\mathbf{B}$ and hyperspectral pixels $\mathbf{Y}$.
- Build the edge set E of the vertices sets $\mathcal{B}$ and $\mathcal{Y}$ in which the sparse representation coefficients $\mathbf{X}$ in Eq. 1 encode the association information between $\mathcal{B}$ and $\mathcal{Y}$.

3: Compute the non-negative matrix $\mathbf{A} = |\mathbf{X}|$.

4: Apply SVD to the normalized matrix $\tilde{\mathbf{A}} = \mathbf{D}_1^{-\frac{1}{2}}\mathbf{A}\mathbf{D}_2^{-\frac{1}{2}}$.

5: Construct the matrix $\mathbf{F} = [\mathbf{D}_1^{-1/2}\mathbf{V}_1 \quad \mathbf{D}_2^{-1/2}\mathbf{V}_2]^T$, where $\mathbf{V}_1$ and $\mathbf{V}_2$ are the left and right singular vectors of normalized matrix $\tilde{\mathbf{A}}$.

6: Run k-means on the rows of $\mathbf{F}$.

Output: Clustering matrix for $\mathbf{Y}$ which records the labels of clustering result.

As shown in Algorithm 1, to construct the bipartite graph G, sparse representation and dictionary atoms are first learned with (1) under the $p = 0$ constraint; the corresponding computational complexity is $O(t(cdn + ndm^2 + nd^2m))$, where n denotes the number of hyperspectral pixels, m denotes the number of dictionary atoms, d denotes the number of bands, c denotes the sparsity degree, and t denotes the number of iterations. In the bipartite graph partition, the computational complexity of SVD for the normalized matrix is $O(m^2n)$, and the computational complexity of k-means is $O(mnkt)$, where k denotes the number of clusters. In conclusion, the whole bipartite graph partition algorithm has a complexity of $O((t(cdn + ndm^2 + nd^2m)) + m^2n + mnkt)$.

In general, HSIs provide detailed spatial land coverage information. The rationale is that the neighboring hyperspectral pixels within the homogeneous spatial area are assumed to represent similar spectral characteristics and should possess similar sparse coefficients. However, in (1), the sparse coefficients are represented as independent, and the relationship among hyperspectral pixels is ignored, which leads to inaccurate coclustering results. To further consider the combination of spectral bands and spatial context information for HSIs, we introduce shape-adaptive superpixel neighborhood information to improve the feature stability and clustering performance. Here, each superpixel can also be called a homogeneous region, which is composed of a series of spatial neighboring pixels. Generally, it is formed by the dictionary atoms of the same class and can be further assumed to be in a low-dimensional subspace.

III. Proposed Method

A. Framework Overview

The proposed coclustering framework for HSIs is shown in Fig. 1. To improve the performance of feature learning and clustering accuracy for large HSIs, we propose a novel bipartite graph partition-based coclustering framework between superpixels and pixels, in which superpixel segmentation is first conducted to learn the dictionary atoms with joint sparsity constraints, followed by a graph NMF step to form a bipartite graph partition. In particular, structured dictionary learning is proposed to express the intrinsic features of superpixel regions as well as construct the vertex sets of bipartite graphs, and graph NMF is further applied to tackle the clustering problem of large HSIs by considering three factors: the low computational complexity, low memory requirement, and the good intrinsic geometry preserving property of an HSI.

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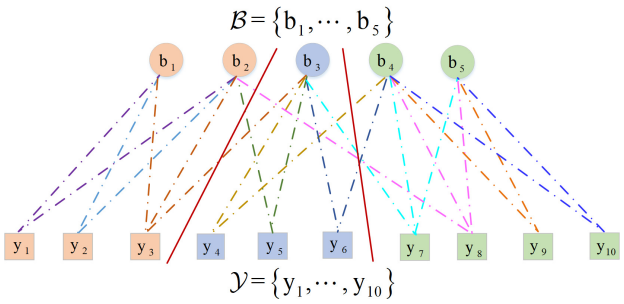


Fig. 2. Bipartite graph consisting of two disjoint sets with 15 vertices that include five superpixel-based dictionary atoms and ten hyperspectral pixels, and the weighted edge linking the disjoint sets are constructed with the sparse representation coefficients. The graph NMF approach solves the coclustering problem of HSIs by grouping pixels with atoms, which utilizes the indicator matrix to find the smallest significant weights between pixels and atoms and preserves the geometric structure of HSIs simultaneously. The solution of graph partitioning is shown by the solid red line. The vertices of the first cluster $\mathcal{V}_1$ are $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$, the vertices of the second cluster $\mathcal{V}_2$ are $\{\mathbf{b}_3, \mathbf{y}_4, \mathbf{y}_5, \mathbf{y}_6\}$, and the vertices of the third cluster $\mathcal{V}_3$ are $\{\mathbf{b}_4, \mathbf{b}_5, \mathbf{y}_7, \mathbf{y}_8, \mathbf{y}_9, \mathbf{y}_{10}\}$.

Given n unlabeled hyperspectral pixels $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_i, \ldots, \mathbf{y}_n] \in R^{d\times n}$, the purpose of the presented coclustering method is to cluster the hyperspectral pixels into discrete groups with unsupervised learned features by constructing a bipartite graph with two sets: pixels and superpixels. In this framework, we first learn the structured dictionary atoms to explore the intrinsic features within superpixel regions and then incorporate the corresponding sparse coefficient matrix into the affinity matrix between dictionary atoms and hyperspectral pixels, as shown in Fig. 2. Then, we can create a weighted bipartite graph $G = (\mathcal{B}, \mathcal{Y}, E)$, where the hyperspectral pixels and dictionary atoms are defined as the disjoint vertices sets $\mathcal{B}$ and $\mathcal{Y}$, respectively. The adjacency matrix for the bipartite graph model is constructed by transforming the joint sparsity representation into a nonnegative affinity matrix. Specifically, to explore the underlying contextual information and local similarities in HSIs, an efficient oversegmentation algorithm is proposed to adaptively adjust the size and shape of superpixels according to the complex spatial contextual structures of HSIs. Intuitively, the pixels within each superpixel have similar spatial structures, and hyperspectral pixels in a superpixel area are usually located in a low-dimensional subspace and have similar sparse coefficients. Therefore, based on the superpixels, we design an optimizing framework for dictionary learning under the joint sparsity constraint to explore the intrinsic local self-similarity and local group sparsity of HSIs simultaneously. For disjoint sets of dictionary atoms and hyperspectral pixels, it can be boiled down to a bipartite graph partition with representations to model the association information among dictionary atoms, joint sparsity representations, and hyperspectral pixels, as shown in Fig. 3. To overcome the bottleneck of computational and memory space in the bipartite graph partition problem for large HSIs, we take advantage of the equivalence of NMF and Laplacian-based spectral clustering and propose an alternative bipartite graph partition with NMF for coclustering of large HSIs to significantly reduce the computational load and memory space compared with traditional Laplacian-based spectral clustering methods.

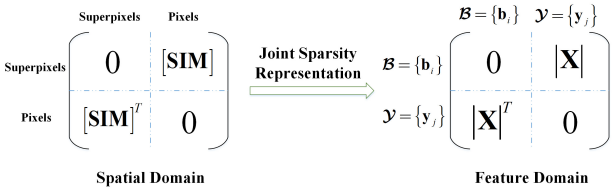


Fig. 3. Bipartite graph constructed from the spatial domain to the feature domain for HSIs. *[SIM] denotes the similarity-based affinity matrix between superpixels and pixels.

In terms of geometry, hyperspectral data are usually sampled from a low-dimensional manifold embedded in a high-dimensional ambient space. Therefore, we incorporate graph regularization into NMF, called graph NMF, to capture the geometric contextual characteristics in HSIs and find a compact representation for the indicator matrix of the adjacency matrix (see Fig. 2 for details). However, the graph Laplacian matrix has a bottleneck with respect to memory space when addressing large HSIs. Inspired by the anchor-based strategy [37], we adopt the superpixel-based anchor graph Laplacian to promote the manifold structure-preserving ability and the burden of memory space for the large hyperspectral data.

B. Superpixel-Based Dictionary Learning

Since HSIs have complex spatial contextual characteristics, effective superpixel segmentation for HSIs that can acquire a window with adjustable sizes can be beneficial for obtaining intrinsic dictionary atoms and better similarity graphs. In this article, the superpixel segmentation-based joint sparsity representation is proposed to learn structured dictionary atoms from both the spatial contextual and spectral characteristics of HSIs. The main steps of dictionary atom learning for superpixels are summarized as follows: 1) superpixel segmentation for HSIs and 2) dictionary and representation learning for superpixels.

1) Superpixel Segmentation for HSIs: To segment HSIs, we first apply principal component analysis to get the first three principal components. After that, they are used to create a superpixel segmentation map via entropy rate superpixel segmentation [38]. Specifically, the three principal components are mapped to an image, and it is represented as a graph $G_1 = (V_1, E_1)$, where V_1 denotes the set of vertices and E_1 is the edge weight set that records the pairwise similarities between vertices. The goal of segmentation is to partition the graph into several connected subgraphs. To group nearby pixels into superpixels with compact, homogeneous, and balanced sizes, the entropy rate constraint is described as follows:

$$\max_{\mathbf{S}} \mathcal{H}(\mathbf{S}) + \alpha\mathcal{P}(\mathbf{S}) \quad \text{s.t. } \mathbf{S} \subseteq E_1 \tag{7}$$

where $\mathbf{S}$ is a subset of E_1, $\mathcal{H}(\mathbf{S})$ denotes the entropy rate term, which is used to form compact and homogeneous clusters, $\mathcal{P}(\mathbf{S})$ denotes the balancing function that promotes clusters with balanced sizes, and α denotes the weight parameter that controls the tradeoff between the entropy rate term and balance term. Equation (7) can be solved with a greedy algorithm [39]. Therefore, the final segmented superpixels can be obtained with the guidance of the segmentation map.

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2) Dictionary and Representation Learning for Superpixels: Using the entropy rate-based optimizing superpixel segmentation for HSIs, we obtain a number of spatial neighborhoods grouped by a superpixel with adaptive shape and size. Let $\{\mathbf{S}_1, \ldots, \mathbf{S}_i, \ldots, \mathbf{S}_w\}$ denote the set of superpixels and $\mathbf{Y}_{\mathbf{S}_i} = [\mathbf{y}_{i,1}, \ldots, \mathbf{y}_{i,|\mathbf{S}_i|}]$ denote a member of superpixel $\mathbf{S}_i$, where w denotes the number of superpixels and $\mathbf{y}_{i,|\mathbf{S}_i|}$ denotes the $|\mathbf{S}_i|$th hyperspectral pixel in the superpixel $\mathbf{S}_i$. In general, hyperspectral pixels within a superpixel $\mathbf{S}_i$ are assumed to be local homogeneous regions, and they can be expressed as a linear combinations of common pattern atoms from the dictionary. By imposing a sparsity regularization on the pixels of each superpixel, we build the joint sparse representation model as follows:

$$\arg\min_{\mathbf{B},\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{B}\mathbf{X}\|_F^2 + \beta \sum_{i=1}^{w} \left\|\mathbf{X}_{\mathbf{S}_i}\right\|_{2,1} \tag{8}$$

where $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_i, \ldots, \mathbf{x}_n] \in R^{m\times n}$ denotes the sparse coefficient from the dictionary $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_i, \ldots, \mathbf{b}_m] \in R^{d\times m}$, $\|\mathbf{X}_{\mathbf{S}_i}\|_{2,1}$ denotes the sum of the l_2 norm of the rows of $\mathbf{X}_{\mathbf{S}_i}$, m denotes the number of dictionary atoms, and β denotes the tradeoff parameter between the reconstruction error and priori. The rationales behind the superpixel-based dictionary and representation learning model are that the hyperspectral pixels $\mathbf{Y}$ can be assumed from a union of low-dimensional subspaces, and $\mathbf{B}$ is the bases of the subspaces. In this end, once the dictionary is learned, each one of the pixels can be represented by a linear combination of few atoms. Each combination of atoms would define a low-dimensional subspace, and thus, the pixels spanned by the same subspace would be represented by similar groups of atoms. The optimization strategy for solving (8) is proposed in [40] and first initializes the dictionary atoms and alternates updating between the dictionary atoms and the sparse coding stage until convergence. Notably, the joint sparsity representation $\mathbf{X}$ is the nonnegative matrix.

C. Bipartite Graph Partition With NMF

From (8), we observe that the locations of the nonzero coefficients represent the dictionary atoms of each signal, and their absolute values represent the corresponding weights of the dictionary atoms. It can be regarded as a bipartite graph $G = (\mathcal{B}, \mathcal{Y}, E)$, where vertices sets $\mathcal{B}$ and $\mathcal{Y}$ denote dictionary atoms and hyperspectral pixels, respectively, and the edge weights E encode the association information between the pairwise elements in both sets of vertices, and the particular associations are encoded by the corresponding sparse representation coefficients.

It is obvious that the bipartite graph partition with the minimum weight of crossing edges between partition regions corresponds to hyperspetral pixel and dictionary atom clustering [26]. However, the classical bipartite graph partition technique in Algorithm 1 is not the first choice for large and high-dimensional hyperspectral datasets since it requires the SVD calculation of the normalized similarity matrix $\mathbf{D}_1^{-(1/2)}\mathbf{A}\mathbf{D}_2^{-(1/2)}$, which requires a large amount of time and memory consumption. In addition, centroid-based clustering methods are usually applied for the obtained singular vectors to get the final clustering results. To overcome these bottlenecks, inspired by NMF, we propose an alternative bipartite graph partition with NMF to significantly reduce the computational and memory load and avoid running the k-means to obtain the final clustering result.

The bipartite graph partition simultaneously groups dictionary atoms and hyperspectral pixels by transforming the sparsity coefficients into an adjacency matrix. Here, the nonnegative joint sparsity coefficient $\mathbf{X}$ contains the association among row and column objects. In this article, we use kernel k-means clustering [33] to partition the bipartite graph $G = (\mathcal{B}, \mathcal{Y}, E)$. Specifically, since the rows of $\mathbf{X}$ are used to encode the associations related to the dictionary atoms $\mathbf{B}$, we define $\mathbf{F} = [\mathbf{f}_1, \ldots, \mathbf{f}_k] \in R^{m\times k}$ as the indicator matrix of the row clusters, where $\mathbf{f}_k$ denotes the probability of all dictionary atoms associated with the kth cluster and $\mathbf{F}_{ik}$ is the (i,k)-element of $\mathbf{F}$ and it also denotes the probability of the ith dictionary atom belonging to the kth cluster. Similarly, we also define $\mathbf{G} = [\mathbf{g}_1, \ldots, \mathbf{g}_k] \in R^{n\times k}$ as the indicator matrix of the column clusters since the columns of $\mathbf{X}$ contain the associations related to the hyperspectral pixels, where $\mathbf{g}_k$ denotes the probability of all hyperspectral pixels associated with the kth cluster, and $\mathbf{G}_{jk}$ denotes the probability of the jth hyperspectral pixel belonging to the kth cluster. The objective function of the bipartite graph partitioned by the kernel k-means, i.e., simultaneous row and column kernel k-means clustering of $\mathbf{X}$, can be described as follows:

$$\begin{cases} \max\limits_{\mathbf{F}^T\mathbf{F}=\mathbf{I},\mathbf{F}\geq 0} \mathbf{Tr}\left(\mathbf{F}^T\mathbf{X}\mathbf{X}^T\mathbf{F}\right) \cdots\cdots \text{Row clustering} \\ \max\limits_{\mathbf{G}^T\mathbf{G}=\mathbf{I},\mathbf{G}\geq 0} \mathbf{Tr}\left(\mathbf{G}^T\mathbf{X}^T\mathbf{X}\mathbf{G}\right) \cdots\cdots \text{Column clustering.} \end{cases} \tag{9}$$

Combining the two equations in (9), we can further rewrite the equation as follows:

$$\begin{aligned} &\max_{\mathbf{F},\mathbf{G}} \mathbf{Tr}\begin{pmatrix}\mathbf{F}\\ \mathbf{G}\end{pmatrix}^T \begin{pmatrix}\mathbf{0} & \mathbf{X}\\ \mathbf{X}^T & \mathbf{0}\end{pmatrix}\begin{pmatrix}\mathbf{F}\\ \mathbf{G}\end{pmatrix} = 2\mathbf{Tr}\left(\mathbf{F}^T\mathbf{X}\mathbf{G}\right) \\ &\text{s.t. } \mathbf{F}^T\mathbf{F} = \mathbf{I}, \quad \mathbf{G}^T\mathbf{G} = \mathbf{I}, \ \mathbf{F},\mathbf{G} \geq 0. \end{aligned} \tag{10}$$

For (10), it is easily observed that this is an objective function of bipartite graph partition type. Next, we illustrate that (9) is equivalent to (10). First, the solution for (10) is achieved by the first k eigenvectors of $\begin{pmatrix}0 & \mathbf{X}\\ \mathbf{X}^T & 0\end{pmatrix}\begin{pmatrix}\mathbf{f}_k\\ \mathbf{g}_k\end{pmatrix} = \lambda_k\begin{pmatrix}\mathbf{f}_k\\ \mathbf{g}_k\end{pmatrix}$, which can be further rewritten as $\mathbf{X}\mathbf{g}_k = \lambda_k\mathbf{f}_k$ and $\mathbf{X}^T\mathbf{f}_k = \lambda_k\mathbf{g}_k$. Multiplying both sides of $\mathbf{X}\mathbf{g}_k = \lambda_k\mathbf{f}_k$ and $\mathbf{X}^T\mathbf{f}_k = \lambda_k\mathbf{g}_k$ by $\mathbf{X}^T$ and $\mathbf{X}$, respectively, we can obtain $\mathbf{X}^T\mathbf{X}\mathbf{g}_k = \lambda_k\mathbf{X}^T\mathbf{f}_k = \lambda_k\mathbf{g}_k$ and $\mathbf{X}\mathbf{X}^T\mathbf{f}_k = \lambda_k\mathbf{X}\mathbf{g}_k = \lambda_k\mathbf{f}_k$, which are also the solutions for the equations in (9). Therefore, we prove that (9) is equivalent to (10).

Furthermore, we can relax (10) into a matrix decomposition problem in which two terms $\mathbf{Tr}(\mathbf{X}^T\mathbf{X})$ and $\mathbf{Tr}(\mathbf{F}^T\mathbf{F}\mathbf{G}^T\mathbf{G})$ are added. Therefore, it can be described as follows:

$$\begin{aligned} &\min_{\mathbf{F},\mathbf{G}} -2\mathbf{Tr}\left(\mathbf{F}^T\mathbf{X}\mathbf{G}\right) \\ &\quad = \mathbf{Tr}\left(\mathbf{X}^T\mathbf{X}\right) - 2\mathbf{Tr}\left(\mathbf{F}^T\mathbf{X}\mathbf{G}\right) + \mathbf{Tr}\left(\mathbf{F}^T\mathbf{F}\mathbf{G}^T\mathbf{G}\right) \\ &\quad = \left\|\mathbf{X} - \mathbf{F}\mathbf{G}^T\right\|_F^2 \\ &\text{s.t. } \mathbf{F}^T\mathbf{F} = \mathbf{I}, \quad \mathbf{G}^T\mathbf{G} = \mathbf{I}, \ \mathbf{F},\mathbf{G} \geq 0. \end{aligned} \tag{11}$$

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For the low rank $\mathbf{G}$, the cluster label of jth hyperspectral pixel is achieved by $\arg\max_k \mathbf{G}_{jk}$ because the rows of $\mathbf{G}$ can be interpreted as the posterior probability for clustering on hyperspectral pixels [41]. Similarly, the cluster label of the ith dictionary atom is achieved by $\arg\max_k \mathbf{F}_{ik}$. Therefore, it can clearly be observed that the low-rank matrix $\mathbf{G}$ will cluster the hyperspectral pixels and the low-rank matrix $\mathbf{F}$ will cluster the dictionary atoms. In addition, the clustering results can be directly obtained without the centroid-based method.

D. Bipartite Graph Partition With Manifold Regularization

The bipartite graph partition with NMF for HSI coclustering is used to find two sets of basis vectors that are utilized to best approximate the dictionary atoms and hyperspectral pixels. However, the inherent discriminating geometrical structure of the hyperspectral pixels cannot be successfully captured. In the HSIs, the pixels of neighbors have strong similarities. We assume that two pixels $\mathbf{y}_i$ and $\mathbf{y}_j$ are close in the geometric contextual structure of the hyperspectral pixels, and the indicator vectors $\mathbf{G}_{i\bullet}$ and $\mathbf{G}_{j\bullet}$ of the pixels $\mathbf{y}_i$ and $\mathbf{y}_j$ are close to each other. In this article, the Laplacian matrix is introduced into the bipartite graph partition model to explore the property of the local invariant of manifolds. After low-rank matrix decomposition of the HSIs, the low-dimensional space maintains the geometric structure of the high-dimensional space $\mathbf{Y}$. The objective model combining bipartite graph partitioning and manifold regularization is described as follows:

$$\begin{aligned}&\min_{\mathbf{F},\mathbf{G}} \left\|\mathbf{X}-\mathbf{F}\mathbf{G}^T\right\|_{\mathbf{F}}^2+\gamma\,\mathbf{Tr}\left(\mathbf{G}^T\bar{\mathbf{L}}\mathbf{G}\right)\\&\text{s.t. } \mathbf{F}^T\mathbf{F}=\mathbf{I},\quad \mathbf{G}^T\mathbf{G}=\mathbf{I},\ \mathbf{F},\mathbf{G}\geq 0\end{aligned}\tag{12}$$

where $\bar{\mathbf{L}}=\bar{\mathbf{D}}-\bar{\mathbf{W}}$ denotes the Laplacian matrix about $\bar{\mathbf{W}}$, $\bar{\mathbf{W}}$ denotes the similarity matrix about the hyperspectral pixels, and $\bar{\mathbf{D}}=\sum_j \bar{\mathbf{W}}_{ij}$ denotes the diagonal matrix.

From (12), we need the computational complexity of $O(n^2d)$ to construct the Laplacian matrix, where n and d are the number of hyperspectral pixels and bands, respectively. However, the Laplacian matrix usually requires high computational time and large memory space for large HSIs, and it fails for large HSIs. In this article, inspired by anchor graph construction [37], we present a superpixel anchor graph-based manifold regularization to overcome the bottlenecks of computational time and memory space in large hyperspectral datasets.

The superpixel anchor graph construction can be divided into two parts: 1) the anchor for the superpixels in the large hyperspectral pixels is constructed and 2) the relationship matrix between pixels and anchors is designed to depict the similarities among data points. In the first part, we utilize the superpixel patches $\mathbf{S}_w$ to construct the anchors of hyperspectral pixels. Specifically, the superpixel patches can be achieved by entropy rate superpixel segmentation [38], and we make use of the superpixel centers as anchors. Compared with the anchors obtained with the rand selection or k-means method, the proposed anchor construction method utilizes the homogeneous regions between the spectral and spatial HSIs, and it is more representative because it uses the superpixel centers. In the second part, the relationship matrix between pixels and anchors is designed to depict the similarities among data points. Let $\mathbf{U}\in R^{n\times w}$ denote the similarities between pixels and anchors and $\mathbf{s}=[\mathbf{s}_1,\ldots,\mathbf{s}_j,\ldots,\mathbf{s}_w]\in R^{d\times w}$ denote the anchors, where w is the number of anchors. For the pairwise relations between pixels and anchors, the kernel function is adopted, and the distances between pixels $\mathbf{y}_i$ and anchors $\mathbf{s}_j$ can be described as follows:

$$\mathbf{U}_{ij}=\frac{K\left(\mathbf{y}_i,\mathbf{s}_j\right)}{\sum_{o\in\Omega_i}K\left(\mathbf{y}_i,\mathbf{s}_o\right)}\tag{13}$$

where $\Omega_i\subseteq\{1,\ldots,w\}$ is the set of indexes of the nearest anchors of $\mathbf{y}_i$, $K(\mathbf{y}_i,\mathbf{s}_j)=\exp(-\|\mathbf{y}_i-\mathbf{s}_j\|_2^2/2\sigma^2)$ and σ is a scale parameter that is regarded as the average value of all pairwise distances [42] in this article. The kernel function has an advantage that is intrinsically related to the Laplace Beltrami operator on the manifold [43]. Note that in (13), $\mathbf{U}_{ij}\geq 0$ and $\sum_{j=1}^{w}\mathbf{U}_{ij}=1$. Using Markov random walks, Liu *et al.* [44] proved that the adjacency matrix between pixels can be represented as follows:

$$\bar{\mathbf{W}}=\mathbf{U}\Lambda^{-1}\mathbf{U}^T\tag{14}$$

where Λ is a diagonal matrix with $\Lambda_{jj}=\sum_{i=1}^{n}\mathbf{U}_{ij}$. Therefore,

$$\begin{aligned}\bar{\mathbf{D}}&=\sum_j\bar{\mathbf{W}}_{ij}=\mathrm{diag}\left(\bar{\mathbf{W}}\mathbf{1}\right)=\mathrm{diag}\left(\mathbf{U}\Lambda^{-1}\mathbf{U}^T\mathbf{1}\right)\\&=\mathrm{diag}\left(\mathbf{U}\Lambda^{-1}\Lambda\mathbf{1}\right)=\mathrm{diag}(\mathbf{U}\mathbf{1})=\mathbf{I}.\end{aligned}$$

Finally, we can rewrite the (12) as follows:

$$\begin{aligned}&\min_{\mathbf{F},\mathbf{G}}\left\|\mathbf{X}-\mathbf{F}\mathbf{G}^T\right\|_{\mathbf{F}}^2+\gamma\,\mathbf{Tr}\left(\mathbf{G}^T\left(\mathbf{I}-\mathbf{U}\Lambda^{-1}\mathbf{U}^T\right)\mathbf{G}\right)\\&\text{s.t. } \mathbf{F}^T\mathbf{F}=\mathbf{I},\quad \mathbf{G}^T\mathbf{G}=\mathbf{I},\ \mathbf{F},\mathbf{G}\geq 0.\end{aligned}\tag{15}$$

E. Optimization Algorithm for the BGP-GNMF

For (15), the objective function is nonconvex in terms of both $\mathbf{F}$ and $\mathbf{G}$, but it is convex in $\mathbf{F}$ or $\mathbf{G}$ individually. In this article, we use the MURs, which are widely used in solving the problems of NMF [45], to solve (15). We introduce a Lagrangian multiplier Λ_1 to enforce $\mathbf{F}^T\mathbf{F}=\mathbf{I}$ and another Lagrangian multiplier Λ_2 to enforce $\mathbf{G}^T\mathbf{G}=\mathbf{I}$. The Lagrangian function of (15) is

$$\begin{aligned}J(\mathbf{F},\mathbf{G})=&\left\|\mathbf{X}-\mathbf{F}\mathbf{G}^T\right\|_{\mathbf{F}}^2+\gamma\,\mathbf{Tr}\left(\mathbf{G}^T\left(\mathbf{I}-\mathbf{U}\Lambda^{-1}\mathbf{U}^T\right)\mathbf{G}\right)\\&+\mathbf{Tr}\left(\Lambda_1\left(\mathbf{F}^T\mathbf{F}-\mathbf{I}\right)\right)+\mathbf{Tr}\left(\Lambda_2\left(\mathbf{G}^T\mathbf{G}-\mathbf{I}\right)\right)\\&\text{s.t. } \mathbf{F}\geq 0,\quad \mathbf{G}\geq 0.\end{aligned}\tag{16}$$

The derivatives of J with respect to $\mathbf{F}$ and $\mathbf{G}$ are expressed as follows:

$$\begin{aligned}\frac{\partial J}{\partial\mathbf{F}}&=-2\mathbf{X}\mathbf{G}+2\mathbf{F}\mathbf{G}^T\mathbf{G}+2\mathbf{F}\Lambda_1\\\frac{\partial J}{\partial\mathbf{G}}&=-2\mathbf{X}^T\mathbf{F}+2\mathbf{G}\mathbf{F}^T\mathbf{F}+2\gamma\left(\mathbf{I}-\mathbf{U}\Lambda^{-1}\mathbf{U}^T\right)\mathbf{G}+2\mathbf{G}\Lambda_2.\end{aligned}\tag{17}$$

Applying the Karush–Kuhn–Tuckre (KKT) conditions [46] to (17), we can obtain the updated $\mathbf{F}_{ik}$ and $\mathbf{G}_{ik}$ as follows:

$$\mathbf{F}_{ik}\leftarrow\mathbf{F}_{ik}\left(\frac{(\mathbf{X}\mathbf{G})_{ik}}{\left(\mathbf{F}\mathbf{G}^T\mathbf{G}+\mathbf{F}\Lambda_1\right)_{ik}}\right)^{1/2}\tag{18}$$

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$$\mathbf{G}_{jk} \leftarrow \mathbf{G}_{jk}\left(\frac{\left(\mathbf{X}^T\mathbf{F}+\gamma\,\mathbf{U}\Lambda^{-1}\mathbf{U}^T\mathbf{G}\right)_{jk}}{\left(\mathbf{G}\mathbf{F}^T\mathbf{F}+\gamma\,\mathbf{G}+\mathbf{G}\Lambda_2\right)_{jk}}\right)^{1/2}. \tag{19}$$

It is worth noting that once we obtain the indicator matrices $\mathbf{F}$ and $\mathbf{G}$, the cluster labels of ith dictionary atom and jth hyperspectral pixel can be aclculated as $\arg\max_k \mathbf{F}_{ik}$ and $\arg\max_k \mathbf{G}_{jk}$, respectively. Finally, the whole BGP-GNMF algorithm is described in Algorithm 2 in detail.

Algorithm Convergence and Correctness:

Theorem 1: The iterative algorithm in (18) and (19) for the objective function in (15) converges.

Theorem 2: At convergence, the solutions for (18) and (19) satisfy the KKT optimality condition, that is, the algorithm correctly converges to local optima.

The proof for Theorems 1 and 2 is given in the Appendix.

Algorithm 2 BGP-GNMF for Large HSIs

Input: $\mathbf{Y} \in R^{d\times n}$: hyperspectral image, k: the number of clusters.

1: Perform dictionary learning under joint sparsity regularization (in Eq. 8) to obtain the superpixel-based structured dictionary atoms $\mathbf{B}$

2: Construct a weighted bipartite graph $G = (\mathcal{B}, \mathcal{Y}, E)$

- Define disjoint vertices sets $\mathcal{B}$ and $\mathcal{Y}$ by dictionary atoms $\mathbf{B}$ and hyperspectral pixels $\mathbf{Y}$.
- Build the edge set E of the vertices sets $\mathcal{B}$ and $\mathcal{Y}$ in which the sparse representation coefficients $\mathbf{X}$ in Eq. 8 encode the association information between $\mathcal{B}$ and $\mathcal{Y}$.

3: Compute the relationship matrix $\mathbf{U}$ between hyperspectral pixels and anchors, where $\mathbf{U}$ is calculated by Eq. 13.

4: Repeat

- Update the indicator matrix $\mathbf{F}$ of dictionary atoms by Eq. 18.
- Update the indicator matrix $\mathbf{G}$ of hyperspectral pi -xels by Eq. 19.

Until the stopping criterion is satisfied

5: Compute the cluster label of jth hyperspectral pixels by $\arg\max_k \mathbf{G}_{jk}$.

Output: Clustering matrix for $\mathbf{Y}$ which records the labels of clustering result.

F. Computational Complexity

The computational complexity of the proposed BGP-GNMF is illustrated in this section. As shown in Algorithm 2, superpixel segmentation via entropy rate superpixel segmentation is first proposed to obtain superpixels with adaptive shapes and sizes for HSIs. The lazy greedy [39] algorithm is introduced to solve this segmentation problem, and the computational complexity is $O(n^2\log(n))$, where n denotes the number of hyperspectral pixels. Then, the joint sparsity representation is learned to construct the affinity matrix, and the processes of sparse representation and dictionary atom learning have a complexity of $O(wt(9m^3+3m^2n+mn))$ [47], where w denotes the number of superpixel patches, t denotes the number of iterations, and m denotes the number of dictionary atoms. Finally, the graph regularization is combined into the bipartite graph partition. The proposed optimization problem is solved by iteratively updating the indicator matrices $\mathbf{F}$ and $\mathbf{G}$ until convergence. The complexity of $\mathbf{F}$ is $O(mnk+2mk^2+nk^2+3mk)$, where k denotes the number of clusters. The complexity of $\mathbf{G}$ is $O(2nmk+nwk+nw^2+mk^2+2nk^2)$, and the complexities for computing $\mathbf{U}$ and Λ^{-1} are $O(onw)$ and $O(nw+w)$, respectively, where o denotes the number of nearest anchors of the hyperspectral pixel. In conclusion, the complexity of updating $\mathbf{F}$ and $\mathbf{G}$ is $O(t(3mnk+mk^2+3nk^2+nwk+nw^2))$.

IV. Experimental Results and Discussion

In this section, we evaluate the effectiveness and efficiency of the proposed method on three HSIs. Our proposed BGP-GNMF is compared with k-means [11], FCM [10], FCM_S1 [48], Ncut [49], USPEC [50], SGCNR [37], RSC [51], SSSC [20], S^5C [52], BGP-JSDL [47], and bipartite graph partition with dictionary learning (BGP-DL) (in Algorithm 1).

A. Experimental Setups

1) Comparison Algorithms: To evaluate the superiority of the proposed method, 11 benchmark clustering methods for HSIs are considered as follows.

k-means [11] finds clusters by iteratively updating the centroids of clusters and assigning the pixels to one of the clustering centroids in the feature space.

FCM [10] groups pixels into clusters by introducing the fuzziness of the belongingness of image pixels. A pixel belongs to a cluster when a high degree of connection between the pixel and cluster exists, and a pixel belongs far away from a cluster when a low degree of connection between the pixel and cluster exists.

FCM_S1 [48] is the FCM with spatial constraints. It illustrates the fuzziness of the belongingness of each pixel and explores the spatial contextual information of the image.

Ncut [49] is the classical spectral clustering method, and it first constructs the similarity matrix and then applies the centroid-based clustering method for the eigenvector of the Laplacian matrix of the similarity graph.

USPEC [50] is scalable and robust spectral clustering and can be divided into two steps. It first constructs a set of base clusterings by ultrascalable spectral clustering and then combines multiple base clusterings into the consensus clustering with more robustness.

SGCNR [37] first constructs the similarity matrix with an anchor graph, and the nonnegative relaxation is added to constrain the cluster indicator matrix.

RSC [51] uses the l_2-norm graph-based on the intrasubspace projection dominance of the l_2-norm-based projection space to construct a sparse similarity graph.

SSSC [20] is an out-of-sample extension of SSC. It first chooses a small number of samples as in-sample data and performs SSC over in-sample data and then encodes out-of-sample data as a linear combination of in-sample data and assigns the nonsampled data to the cluster.

S^5C [52] selects subsamples based on the approximated subgradients to construct the similarity matrix and solves spectral

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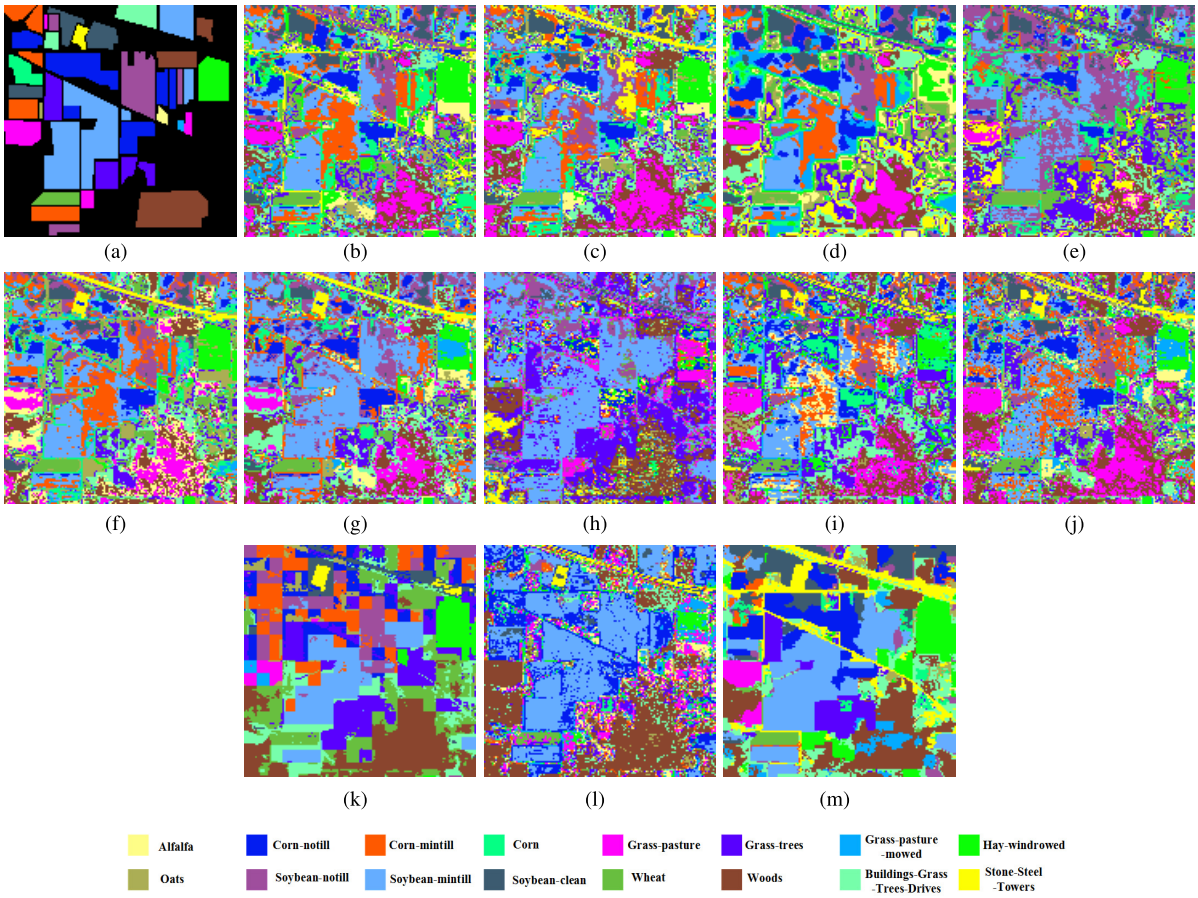


Fig. 4. Clustering results of Indian Pines image. (a) Ground truth. (b) k-means method. (c) FCM method. (d) FCM_S1 method. (e) Ncut method. (f) USPEC method. (g) SGCNR method. (h) RSC method. (i) SSSC method. (j) S^5C method. (k) BGP-JSDL method. (l) BGP-DL method. (m) BGP-GNMF method.

clustering step by performing eigenvalue decomposition using orthogonal iteration.

BGP-JSDL [47] uses a bipartite graph partition-based classifier to simultaneously group hyperspectral pixels with the learned dictionary atoms by mapping the joint sparsity coefficients to a nonnegative adjacency matrix of graph.

BGP-DL (in Algorithm 1) is a coclustering clustering method for hyperspectral pixels and dictionary atoms. First, the dictionary atoms are achieved by K-SVD, and the coefficients are mapped into the adjacency matrix of the bipartite graph. Then, Ncut is used for the adjacency matrix to get the clustering result of hyperspectral pixels and dictionary atoms.

In all the above methods, the number of clusters is set referencing the ground truth in each dataset. For graph-based clustering methods, such as Ncut, we utilize the self-tuned Gaussian method to construct the similarity graph. For centroid-based clustering methods, such as FCM, FCM_S1, and k-means, the initial selection of centroids is very sensitive; thus, we repeatedly run and report the optimal performance in this article. In all clustering methods, the parameters are manually determined to the optimum.

2) Evaluation Metrices: The famous evaluation metrics for different clustering methods in HSIs are used and include the overall accuracy (OA), the user's accuracy (UA), and the produce's accuracy (PA). Specifically, the OA is the percentage of the total number of pixels correctly classified to the total number of reference pixels in HSIs. The UA is number of the correctly classified pixels in each category divided by the total number of pixels that were assigned to that category. The PA is the percentage of correctly classified pixels in each category to the total number of pixels in the corresponding category in the image. Furthermore, normalized mutual information (NMI), adjusted mutual information (AMI), adjusted Rand index (ARI), Fowlkes–Mallows index (FMI), V-measure, completeness, homogeneity, and accuracy (ACC), which have different working mechanisms, are applied in the HSIs to validate the clustering performance of different methods.

B. Dataset Descriptions and Experimental Analysis

1) Indian Pines Image: The Indian Pines image was achieved by an airborne visible infrared imaging spectrometer (AVIRIS) spectrograph in the Indian Pines area of Northwest Indiana. The original ground truth of the Indian Pines image has a size of 145 $\times$ 145 and contains 16 different land-cover classes. To obtain a better clustering performance, some bands with high noise and water absorption are removed, and the spectral bands are reduced from 220 to 200. The similar and mixed spectral features make the clustering of Indian Pines images a challenging task. The maps of the clustering results achieved by different clustering methods are shown in Fig. 4, and the corresponding clustering accuracies are given in detail in Tables I and II. In Tables I and II, the bold

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TABLE I

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE INDIAN PINES IMAGE [PA, UA, AND OA (%)]

Method	Class	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSSC	S^5C	BGP-JSDL	BGP-DL	BGP-GNMF
Produce's	1	11.11	20.37	**48.15**	0	0	18.52	11.11	1.85	11.11	0	38.89	0
accuracy	2	26.22	26.64	39.54	19.53	29.43	30.96	13.81	21.97	32.78	15.41	26.71	**48.61**
(PA)	3	35.37	27.10	26.86	0.48	**40.89**	3.12	4.20	34.65	32.61	31.18	0.60	5.04
(%)	4	41.45	28.21	36.32	**61.11**	19.23	20.09	8.55	32.48	8.97	0	14.10	26.92
	5	48.69	58.15	53.72	31.79	39.24	44.87	18.11	60.36	62.78	40.85	12.68	**64.39**
	6	28.92	28.25	27.98	42.17	30.39	45.25	74.56	55.15	59.84	91.83	27.71	**92.37**
	7	0	65.38	0	0	0	0	0	50.00	76.92	0	**76.92**	0
	8	98.36	93.05	60.33	86.71	93.25	60.53	22.70	53.99	56.03	99.18	38.04	**99.39**
	9	0	10.00	35.00	0	55.00	70.00	0	0	20.00	**90.00**	0	55.00
	10	38.84	29.96	42.67	**55.06**	31.92	33.26	22.83	27.07	28.51	45.04	2.69	3.20
	11	38.01	36.87	30.15	49.96	40.44	67.46	71.76	36.02	41.69	52.15	**78.53**	71.15
	12	29.97	23.62	31.60	16.94	18.73	13.19	7.17	11.40	18.24	26.22	9.45	**45.28**
	13	90.57	95.28	56.60	99.53	95.75	95.28	7.55	77.83	83.49	95.75	68.40	**99.53**
	14	37.64	44.67	36.86	40.03	33.38	42.66	63.52	47.76	46.14	74.11	**82.23**	72.57
	15	15.79	18.42	24.21	21.32	20.79	23.95	3.42	26.05	8.16	11.05	12.11	**26.58**
	16	0	42.11	0	0	87.37	88.42	1.05	67.37	83.16	98.95	72.63	**100**
User's	1	5.94	5.67	6.34	0	0	**14.29**	2.75	0.12	3.61	0	8.61	0
accuracy	2	45.63	52.26	44.26	**64.81**	56.27	56.63	47.48	37.37	49.84	24.89	24.66	42.24
(UA)	3	18.19	21.18	17.81	1.79	18.58	5.37	11.25	21.86	15.64	16.91	4.72	**28.38**
(%)	4	12.37	10.58	21.09	21.70	11.60	5.77	23.81	10.26	9.21	0	27.27	**34.81**
	5	34.47	29.46	33.93	44.38	30.09	31.81	23.68	34.68	30.32	100	15.91	**100**
	6	69.01	76.45	80.08	61.17	82.25	81.06	31.38	57.54	76.28	63.28	65.51	**95.04**
	7	0	**10.43**	0	0	0	0	0	8.02	7.38	0	5.59	0
	8	62.39	89.22	57.06	**91.77**	88.72	87.57	35.46	54.21	82.53	91.17	71.81	69.43
	9	0	0.64	2.15	0	2.86	8.00	0	0	2.19	13.33	0	**15.94**
	10	**56.54**	25.28	52.68	19.60	35.44	22.56	24.42	34.25	34.72	31.85	24.76	21.23
	11	47.33	51.41	59.66	54.25	48.90	49.09	43.25	52.02	48.47	**67.63**	43.48	54.94
	12	25.92	27.88	25.53	23.27	25.61	27.27	39.29	22.29	24.14	**59.63**	32.40	22.35
	13	32.76	42.89	29.56	25.54	33.50	40.81	6.48	70.82	49.30	31.57	61.70	**99.53**
	14	85.59	80.06	91.73	93.67	85.21	88.32	69.96	84.89	84.08	**100**	68.87	90.55
	15	20.07	28.69	25.84	16.56	34.05	17.17	28.89	23.63	19.62	6.80	16.91	**60.12**
	16	0	6.32	0	0	91.21	82.35	0.98	37.87	28.21	94.95	**95.83**	36.40
OA		38.11	37.57	35.89	38.63	37.78	42.40	37.67	37.02	39.81	48.77	41.13	**55.18**

1. Alfalfa, 2. Corn-notill, 3. Corn-mintill, 4. Corn, 5. Grass-pasture, 6. Grass-trees, 7. Grass-pasture-mowed, 8. Hay-windrowed, 9. Oats, 10. Soybean-notill, 11. Soybean-mintill, 12. Soybean-clean, 13. Wheat, 14. Woods, 15. Buildings-Grass-Trees-Drives, 16. Stone-Steel-Towers.

TABLE II

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE INDIAN PINES IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSSC	S^5C	BGP-JSDL	BGP-DL	BGP-GNMF
NMI	43.86	43.80	42.17	43.74	44.70	46.36	29.72	39.03	43.05	49.65	41.67	**56.16**
AMI	42.17	41.33	39.43	42.99	42.87	45.83	27.18	38.75	42.78	49.45	41.36	**54.17**
ARI	21.95	22.95	20.70	22.12	22.48	26.45	20.14	20.24	23.54	30.57	26.81	**35.70**
FMI	30.71	31.25	29.02	32.47	31.28	36.64	33.76	28.83	32.40	38.70	40.72	**44.93**
V-meature	43.84	43.74	42.10	43.74	44.67	46.35	29.64	39.03	43.05	49.65	41.67	**56.13**
Completeness	42.43	41.59	39.69	44.23	43.12	46.62	32.10	37.37	41.93	49.46	45.72	**58.01**
Homogeneity	45.35	46.13	44.81	43.26	46.34	46.09	27.53	40.85	44.24	49.85	38.28	**54.38**
ACC	38.11	37.57	35.89	38.63	37.78	42.40	37.67	37.02	39.81	48.77	41.13	**55.18**

and underlined sections of each row indicate the optimal and the second best clustering accuracy, respectively.

The accuracy of the different clustering methods in Table I illustrates that the proposed BGP-GNMF method is more effective than the other clustering methods. Among the clustering results obtained by centroid-based clustering, the k-means method obtains the best clustering performance compared with FCM and FCM_S1, especially for FCM_S1. FCM_S1 has poor clustering results and contains many misclassifications because the Indian Pines image has similar and mixed spectral signatures. For the clustering performance acquired by the graph-based methods, SGCNR achieves better clustering results than Ncut, USPEC, RSC, SSSC, and S^5C and achieves improvements in the OA of almost 3.77%, 4.62%, 4.73%, 5.38%, and 2.59%, respectively. Specifically, the clustering performance in terms of the PA is increased by 70.00%, 15.00%, 70.00%, 70.00%, and 50.00% for the Oats class, respectively, and the clustering performance in terms of the PA is increased by 88.42%, 1.05%, 87.37%, 21.05%, and 5.26% for the Stone-Steel-Towers class, respectively. Compared with Ncut, USPEC, RSC, and SSSC, S^5C has higher precision in terms of OA and is improved by 1.18%, 2.03%, 2.14%, and 2.79%, respectively. In the above analysis of centroid- and graph-based clustering methods, the graph-based methods take advantage of the intrinsic similarity relationship between the hyperspectral pixels to promote the improvement in clustering performance and achieve smoother map results. Compared with S^5C and SGCNR, the proposed BGP-GNMF method

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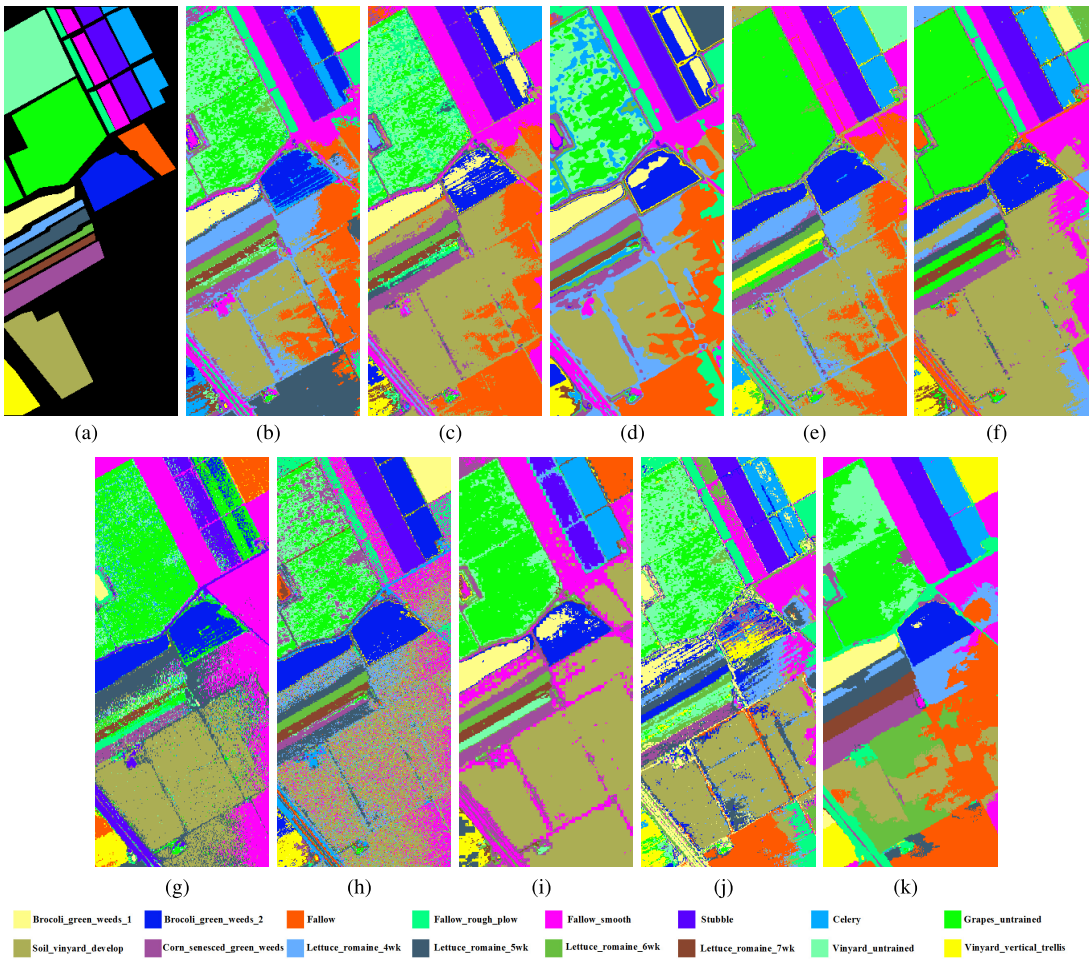


Fig. 5. Clustering results of Salinas image. (a) Ground truth. (b) k-means method. (c) FCM method. (d) FCM_S1 method. (e) USPEC method. (f) SGCNR method. (g) SSSC method. (h) S^5C method. (i) BGP-JSDL method. (j) BGP-DL method. (k) BGP-GNMF method.

obtains superior clustering results. Specifically, the clustering result in terms of the PA is increased by 32.53% and 47.12% for the Grass-trees class, the clustering result in terms of the PA is increased by 43.36% and 38.86% for the Hay-windrowed class, and the clustering result in terms of the PA is increased by 16.84% and 11.58% for the Stone-Steel-Towers class. In the bipartite graph partition-based clustering methods, such as BGP-JSDL and BGP-DL, the proposed BGP-GNMF method has achieved an improvement in terms of the OA of almost 6.41% and 14.05%, respectively. Specifically, from the perspective of the Corn-notill class, the PA is increased by 33.20% and 21.90%. From the perspective of the Grass-pasture class, the PA is increased by 23.54% and 51.71%. Therefore, the proposed BGP-GNMF method is superior to the other methods for HSI clustering both visually and quantitatively and gets the best clustering results with the best OA of 55.18%.

In addition, NMI, AMI, ARI, FMI, V-measure, completeness, homogeneity, and ACC are applied to the Indian Pines image to validate the clustering performance for the proposed BGP-GNMF. From the clustering accuracy of the different evaluation metrics shown in Table II, our BGP-GNMF method obtains better clustering performance compared with the other clustering methods. Specifically, among the clustering performances of the centroid-based methods, FCM_S1 achieves poor clustering accuracy and is not well separated; however, the clustering map of FCM_S1 is much smoother compared with those of FCM and k-means. In the clustering performance of the graph-based methods, SGCNR and S^5C obtain better clustering results. Compared with BGP-DL, BGP-JSDL achieves better clustering results. Compared with SGCNR, S^5C, and BGP-JSDL, the BGP-GNMF proposed in this article gets the best clustering performance based on the visual effects and accuracy. This further illustrates that the proposed BGP-GNMF method can exploit the intrinsic characteristics of the Indian Pines dataset.

2) Salinas Image: The Salinas image was obtained by an AVIRIS spectrograph in the Salinas Valley area of California. The original ground truth of the Salinas image has a size of 512 $\times$ 217 and contains 16 different land-cover classes. In our experiment, we used 204 spectral bands because a total of 20 noisy bands and water absorption (108–112, 154–167, and 224) were removed from the original image. The Salinas image has a high spatial resolution. The maps of the clustering results achieved by the different clustering methods are shown in Fig. 5, and the corresponding clustering accuracies are given in detail in Tables III and IV.

Table III and Fig. 5 show that the proposed BGP-GNMF method is more effective and generates a smoother clustering map in contrast to the other typical clustering methods. For the centroid-based clustering methods, the clustering

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TABLE III

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE SALINAS IMAGE [PA, UA, AND OA (%)]. "-" DENOTES THE OUT OF MEMORY

Method	Class	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSSC	S^5C	BGP-JSDL	BGP-DL	BGP-GNMF
Produce's	1	98.51	99.80	99.05	-	0	0	-	0	0	98.81	65.59	**100**
accuracy	2	83.90	70.29	81.45	-	**98.42**	96.22	-	87.98	97.88	73.78	21.98	98.34
(PA)	3	38.01	15.79	33.60	-	**42.00**	0.10	-	0	4.25	0	1.16	31.28
(%)	4	**99.71**	0	99.07	-	0	99.35	-	0	83.36	0	99.14	0.14
	5	99.22	99.10	93.61	-	96.71	99.10	-	**99.89**	76.89	89.69	65.42	99.78
	6	99.44	99.01	96.74	-	99.39	99.39	-	80.90	99.47	84.87	98.21	**99.80**
	7	38.53	0	0.03	-	99.36	97.46	-	0.36	0.03	99.22	84.91	**99.55**
	8	58.72	42.09	50.94	-	97.18	**98.94**	-	82.97	52.86	86.48	47.64	86.57
	9	94.44	98.15	92.28	-	96.07	**98.34**	-	96.76	64.24	96.05	91.02	63.73
	10	56.35	63.27	45.79	-	59.46	63.79	-	45.15	24.67	62.54	42.56	**95.21**
	11	96.82	0	**99.91**	-	98.31	3.00	-	0	56.46	0	62.55	67.13
	12	0.16	0.21	0	-	45.36	48.26	-	71.67	67.20	0	50.18	**100**
	13	98.03	97.05	95.85	-	97.49	0	-	18.45	**98.14**	97.16	80.13	0
	14	79.53	81.50	88.79	-	0	92.90	-	63.08	89.07	92.80	0	**98.22**
	15	61.53	51.39	54.64	-	0	0	-	18.82	46.62	17.74	**70.46**	62.47
	16	0	43.88	53.02	-	62.48	67.63	-	72.72	66.52	54.62	60.32	**90.20**
User's	1	**99.85**	41.79	48.60	-	0	0	-	0	0	67.91	64.16	90.05
accuracy	2	56.98	55.24	56.37	-	64.65	64.02	-	52.55	38.26	99.13	27.58	**100**
(UA)	3	97.41	85.01	**100**	-	89.25	33.33	-	0	16.00	0	29.11	76.77
(%)	4	98.65	0	95.37	-	0	**99.50**	-	0	72.90	0	98.71	1.55
	5	**78.56**	55.40	73.65	-	59.88	68.30	-	41.61	55.13	44.75	59.53	51.69
	6	99.52	99.90	97.04	-	99.95	99.87	-	100	99.75	**100**	99.56	99.97
	7	48.10	0	0.03	-	86.48	97.98	-	0.73	1.09	99.05	84.16	**99.97**
	8	73.09	77.72	**84.12**	-	58.52	54.78	-	51.27	73.89	62.23	70.09	78.51
	9	97.50	69.69	96.02	-	87.25	74.96	-	96.11	70.69	67.32	98.14	**100**
	10	61.69	55.07	54.29	-	78.05	89.59	-	**95.79**	20.11	64.28	53.99	90.23
	11	33.70	0	30.13	-	50.90	3.62	-	0	24.17	0	60.67	**68.55**
	12	2.73	0.47	0	-	**84.61**	83.11	-	45.94	36.94	0	39.15	82.60
	13	35.48	81.34	46.21	-	48.48	0	-	**86.22**	82.63	65.73	24.36	0
	14	55.84	65.12	74.69	-	0	66.49	-	**94.27**	84.64	81.13	0	53.24
	15	49.94	42.79	50.64	-	0	0	-	36.03	46.32	36.46	43.34	**79.57**
	16	0	83.21	84.33	-	51.98	86.30	-	99.02	85.49	**100**	41.24	99.09
OA		67.99	56.73	63.18	-	69.07	69.41	-	57.10	55.38	66.44	61.50	**79.76**

1. Brocoli_green_weeds_1, 2. Brocoli_green_weeds_2, 3. Fallow, 4. Fallow_rough_plow, 5. Fallow_smooth, 6. Stubble, 7.Celery, 8. Grapes_untrained, 9. Soil_vinyard_develop, 10. Corn_senesced_green_weeds, 11. Lettuce_romaine_4wk, 12. Lettuce_romaine_5wk, 13. Lettuce_romaine_6wk, 14. Lettuce_romaine_7wk, 15. Vinyard_untrained, 16. Vinyard_vertical_trellis.

TABLE IV

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE SALINAS IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSSC	S^5C	BGP-JSDL	BGP-DL	BGP-GNMF
NMI	73.56	68.59	70.89	-	78.91	79.86	-	66.20	64.84	75.23	66.22	**84.68**
AMI	72.62	66.80	70.56	-	72.29	71.57	-	66.17	64.81	75.21	66.19	**84.31**
ARI	54.66	46.93	51.19	-	63.43	59.22	-	45.53	43.21	57.91	48.21	**69.38**
FMI	59.28	52.42	55.96	-	70.39	68.06	-	54.13	49.14	63.85	53.64	**72.55**
V-meature	73.56	68.56	70.89	-	78.61	79.38	-	66.20	64.84	75.23	66.22	**84.68**
Completeness	74.50	70.39	70.59	-	86.12	**89.08**	-	72.06	66.28	79.48	67.16	85.04
Homogeneity	72.64	66.83	71.20	-	72.31	71.59	-	61.22	63.46	71.42	65.30	**84.32**
ACC	67.99	56.73	63.18	-	69.07	69.41	-	57.10	55.38	66.44	61.50	**79.76**

result of FCM contains many significant misclassifications resulting in poor clustering accuracy, especially for the Fallow_rough_plow class, Celery class, Lettuce_romaine_4wk class, and Lettuce_romaine_5wk class. Compared with FCM, k-means and FCM_S1 achieve better clustering results by reducing the number of misclassifications and obtain significant improvement in the OA of almost 11.26% and 6.45%, respectively. Compared with the k-means and FCM, the clustering map of FCM_S1 is much smoother. For graph-based clustering, SGCNR has better clustering performance and achieves a better clustering map than USPEC, SSSC, and S^5C. Specifically, the clustering performance in terms of the PA is increased by 99.35%, 99.35%, and 15.99% for the Fallow_rough_plow class, and the clustering performance in terms of the PA is increased by 92.90%, 29.82%, and 3.83% for the Lettuce_romaine_7wk class. For the clustering accuracy acquired by the centroid- and graph-based clustering methods, the graph-based methods can achieve better clustering performance and smoother clustering maps. Specifically, compared with k-means, SGCNR achieves improvements in the OA of almost 1.42%. Compared with the SGCNR, the proposed BGP-GNMF method improves the clustering accuracy by reducing most of the misclassifications and the OA is improved by 10.35%. Specifically, the clustering performance results in terms of the PA are increased by 100% for the Brocoli_green_weeds_1 class, and the clustering

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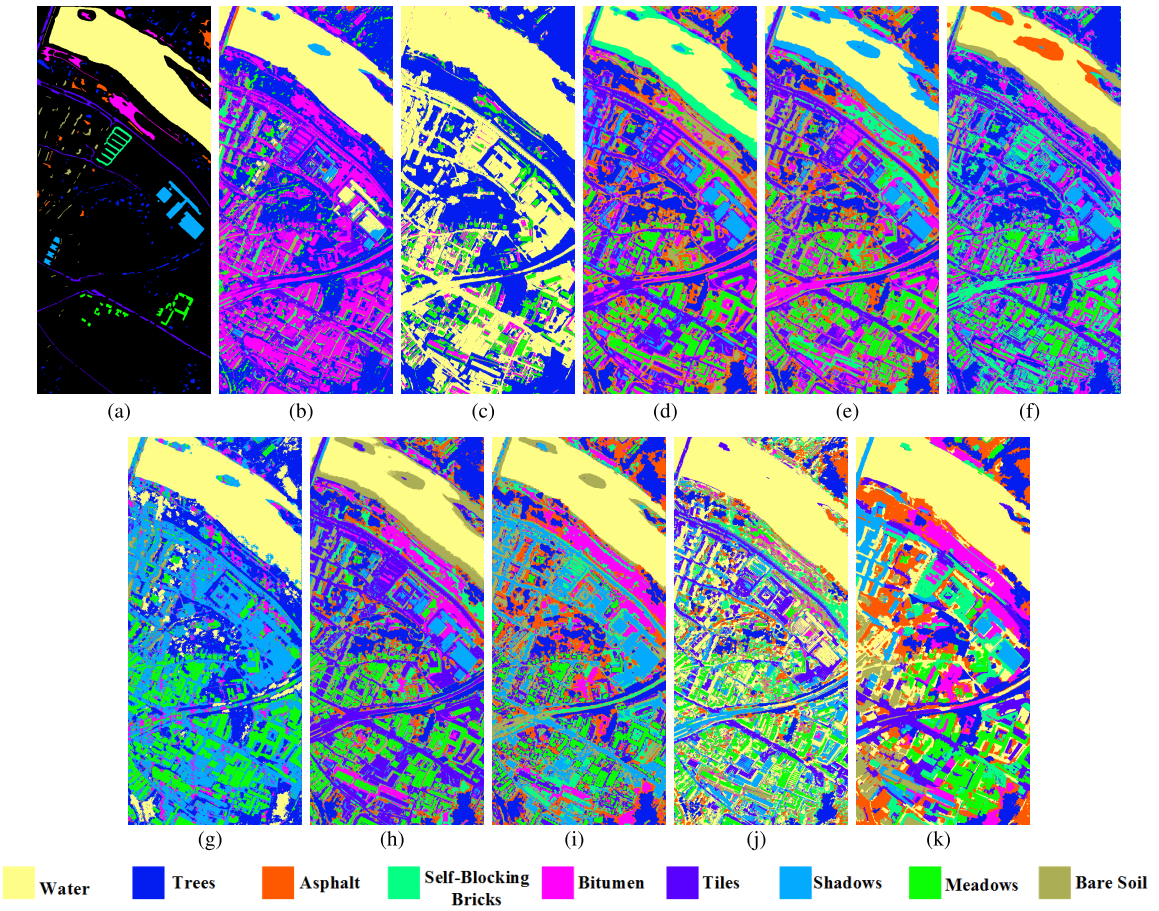


Fig. 6. Clustering results of Pavia Center image. (a) Ground truth. (b) k-means method. (c) FCM method. (d) FCM_S1 method. (e) USPEC method. (f) SGCNR method. (g) SSSC method. (h) S^5C method. (i) BGP-JSDL method. (j) BGP-DL method. (k) BGP-GNMF method.

performance in terms of the PA is increased by 51.74% for the Lettuce_romaine_5wk class. Compared with BGP-JSDL and BGP-DL, the proposed BGP-GNMF method performs better and achieves an improvement in terms of the OA of almost 13.32% and 18.26%, respectively. Specifically, the clustering accuracy in terms of the PA is separately increased by 24.56% and 76.36% for the Brocoli_green_weeds_2 class, and the clustering accuracy in terms of the PA is separately increased by 35.58% and 29.88% for the Vinyard_vertical_trellis class. Therefore, the BGP-GNMF proposed in this article is significantly superior to the other typical methods in HSI clustering both visually and quantitatively and achieves the best clustering accuracy with an OA of 79.76%.

Furthermore, for HSI clustering, the typical scores for evaluation metrics, which are NMI, AMI, ARI, FMI, V-measure, completeness, homogeneity, and ACC, are shown in Table IV. With respect to the clustering accuracy of the different evaluation metrics shown in Table IV, the proposed BGP-GNMF obtains the optimal clustering results among the compared methods. Specifically, the clustering results of FCM demonstrate poor clustering performance, especially for ARI and FMI. Compared with SSSC and S^5C, USPEC and SGCNR successfully identify most land-cover classes to boost the clustering performance and obtain better clustering results. The proposed BGP-GNMF performs much better than USPEC, SGCNR, BGP-JSDL, and BGP-DL in terms of NMI, AMI, ARI, FMI, V-measure, homogeneity, and ACC. In conclusion, these evaluation performances further illustrate the superiority of the proposed BGP-GNMF.

3) Pavia Center Image: The Pavia Center image was obtained by the ROSIS sensor in terms of the Pavia area of Italy. The spectral covering range is 430–860 nm, with 102 spectral bands. However, in this experiment, some bands with high noise and some samples with no information are discarded and removed before the analysis. In this article, we took the test image with a size of 1096 × 490, consisting of nine different land-cover classes. The maps of the clustering results achieved by different clustering methods are shown in Fig. 6, and the corresponding clustering accuracies are given in detail in Tables V and VI.

From Table V and Fig. 6, it can clearly be observed that the proposed BGP-GNMF is more effective compared with the other clustering methods. Among the clustering results obtained by the centroid-based clustering methods, the clustering accuracy of FCM_S1 is much higher than that of the k-means and the FCM because it can successfully identify most of the classes and the OAs improved by 0.95% and 5.33%, respectively. For graph-based clustering, S^5C performs better and achieves a better clustering map than USPEC, SGCNR, and SSSC. Compared with FCM_S1,

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TABLE V

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE PAVIA CENTER IMAGE [PA, UA, AND OA (%)]

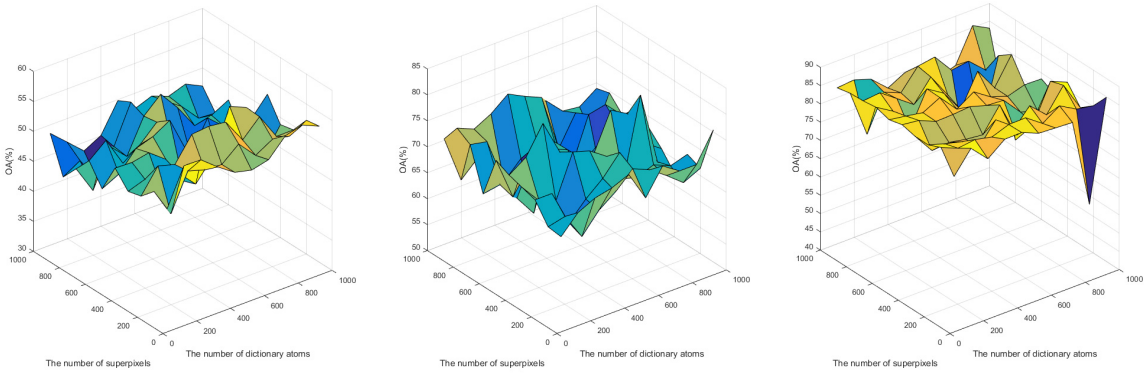
Method	Class	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSSC	S^5C	BGP-JSDL	BGP-DL	BGP-GNMF
Produce's	1	97.50	100	82.99	-	77.19	83.23	-	96.08	85.13	95.07	99.97	**100**
accuracy	2	99.97	**100**	97.23	-	89.33	98.20	-	65.74	90.01	86.86	67.40	72.28
(PA)	3	0	0	34.45	-	55.03	0	-	0	50.16	**73.50**	64.93	67.66
(%)	4	0.05	0	0	-	0	77.66	-	0	0	15.42	0	**81.31**
	5	**64.64**	18.08	45.40	-	58.39	46.95	-	36.01	58.88	49.24	23.96	55.92
	6	43.77	0	89.00	-	97.89	84.00	-	11.56	**99.40**	0.04	83.95	44.09
	7	17.32	0	85.92	-	83.60	84.51	-	**99.30**	84.27	97.87	5.12	73.31
	8	0	75.40	99.33	-	98.33	91.96	-	**99.97**	99.90	85.91	72.68	81.84
	9	42.40	0.32	14.64	-	**85.22**	22.73	-	45.40	47.39	47.85	0	1.06
User's	1	92.85	76.61	100	-	100	100	-	96.01	100	**100**	85.76	94.68
accuracy	2	65.79	52.94	71.32	-	78.56	60.49	-	55.30	79.86	88.20	**92.43**	85.01
(UA)	3	0	0	57.21	-	68.90	0	-	0	**69.15**	56.74	49.27	40.78
(%)	4	0.20	0	0	-	0	27.43	-	0	0	14.45	0	**38.32**
	5	36.25	61.16	89.68	-	82.17	75.48	-	72.91	**99.95**	91.70	96.91	92.10
	6	43.61	0	59.77	-	68.83	86.19	-	**89.51**	55.59	0.56	64.05	72.54
	7	41.03	0	79.49	-	28.97	89.82	-	35.04	**96.41**	43.56	19.87	66.79
	8	0	58.53	70.88	-	99.97	100	-	69.63	97.93	**100**	60.39	97.15
	9	74.57	**100**	25.65	-	99.57	14.41	-	96.47	9.63	17.70	0	4.85
OA		77.19	72.81	78.14	-	77.31	78.58	-	78.84	81.67	81.43	79.30	**85.61**

1. Water, 2. Trees, 3. Asphalt, 4. Self-Blocking Bricks, 5. Bitumen, 6. Tiles, 7. Shadows, 8. Meadows, 9. Bare Soil.

TABLE VI

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE PAVIA CENTER IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSSC	S^5C	BGP-JSDL	BGP-DL	BGP-GNMF
NMI	67.53	54.19	72.63	-	72.43	71.09	-	64.71	**76.06**	75.98	60.62	70.17
AMI	61.96	36.10	67.83	-	67.85	65.26	-	64.71	**76.05**	75.97	60.61	67.64
ARI	79.14	46.91	71.70	-	63.51	71.49	-	79.30	74.48	86.26	69.23	**87.02**
FMI	88.49	77.04	82.71	-	77.32	82.58	-	88.30	84.49	92.41	84.76	**92.82**
V-meature	67.28	50.01	72.46	-	72.28	70.83	-	64.71	**76.06**	75.98	60.62	70.12
Completeness	73.58	**81.31**	67.83	-	67.86	65.27	-	71.52	72.10	76.23	70.25	72.79
Homogeneity	61.97	36.11	77.77	-	77.31	77.44	-	59.09	**80.47**	75.73	53.32	67.65
ACC	77.19	72.81	78.14	-	77.31	78.58	-	78.84	81.67	81.43	79.30	**85.61**



(a) (b) (c)

Fig. 7. Analysis for parameter w and m. (a) Indian Pines image. (b) Salinas image. (c) Pavia Center image.

S^5C can significantly improve the accuracy of clustering and obtains an improvement in terms of OA of nearly 3.53%. Compared with S^5C, the proposed BGP-GNMF significantly decreases most of the misclassifications by improving the OA of almost 3.94%. Compared with BGP-JSDL and BGP-DL, the proposed BGP-GNMF performs better and achieves an improvement in terms of the OA of almost 4.18% and 6.31%, respectively.

In addition, the clustering accuracy evaluation metrics are compared to the typical evaluation metrics in Table VI. Table VI shows that the proposed BGP-GNMF obtains better results and evaluation performances than the other typical clustering methods on the Pavia Center images. Therefore, the evaluation results in Table VI further show the effectiveness of the proposed BGP-GNMF.

C. Parameter Analysis

The proposed BGP-GNMF method totally includes six important parameters that have a great influence on the clustering results. The leave-one-out cross validation is utilized to analyze the sensitivity of each parameter. Figs. 7–11 show the changing trend of the proposed BGP-GNMF algorithm in

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clustering accuracy OA versus each parameter for Indian Pines image, Salinas image, and Pavia Center image.

1) Parameters w and m: Parameter w is the number of superpixels that is obtained via entropy rate superpixel segmentation to create superpixel segmentation map. It determines the quality of contextual information and local similarities in the HSIs. Parameter m is the number of dictionary atoms that is obtained under superpixels and joint sparsity-constrained optimization to better explore the intrinsic feature of the HSIs. Parameters w and m balance the representative ability and the redundancy of the dictionary. Therefore, parameters w and m directly affect the final clustering performance of the proposed BGP-GNMF. The range of the parameters w and m for each of the three HSIs is set from 100 to 1000.

In the changing curves of the Indian Pines dataset shown in Fig. 7(a), the best performance with the highest OA is obtained when the parameter w and parameter m are set to 300 and 500, respectively. In the changing curves of the Salinas dataset shown in Fig. 7(b), the BGP-GNMF method gets the best clustering results and the highest OA when the parameter w and the parameter m are set to 400 and 800, respectively. In the changing curve of the Pavia Center dataset shown in Fig. 7(c), the highest clustering OA result is obtained when the parameter w is set to 1000 and the parameter m is set to 900. Thus, for all three HSIs, in this article, the number of superpixels w can be set to 300, 400, and 1000, and the number of dictionary atoms m can be set to 500, 800, and 900.

2) Parameter α: Parameter α is the weight parameter that controls the tradeoff between the entropy rate term and the balance term. Through α, it can generate compact, homogeneous, and balanced clusters. The range of the parameter α for each of the three HSIs is set from 0.1 to 1. From Fig. 8, with the change of α, the performance of the proposed BGP-GNMF is relatively stable, and it obtains the best performance with the highest OA when parameter α is set to 0.5 for all three hyperspectral datasets.

3) Parameter β: Parameter β is used to balance the reconstruction error and the sparse prior. Through β, we can directly adjust the tradeoff between data fidelity and sparsity of coefficient. The range of the parameter β for each of the three HSIs is set from 1 to 10. From Fig. 9, we can observe that the BGP-GNMF performs good in most cases, and the best performance of the proposed algorithm is obtained when parameter β is set to 3, 2, and 3 for Indian Pines image, Salinas image, and Pavia Center image, respectively.

4) Parameter γ: Parameter γ is utilized to balance the graph partitioning and the manifold regularization. The range of the parameter γ for each of the three HSIs is set from 0 to 10. From Fig. 10, it can be observed that the spatial information plays an important role in the clustering performance, and it obtains the best performance with the highest OA when parameter γ is set to 8, 6, and 5 for Indian Pines image, Salinas image, and Pavia Center image, respectively.

5) Parameter o: Parameter o denotes the number of the nearest anchors of hyperspectral pixels. It is generally related to the spatial distribution of HSI, and it determines the size of the spatial neighborhood. The range of the parameter o for each of the three HSIs is set from 1 to 50. As shown in Fig. 11, the best clustering accuracy with the highest OA is obtained when the parameter is set to 5 for all three hyperspectral datasets.

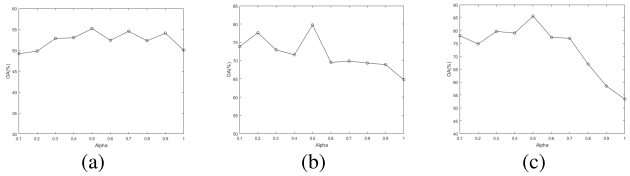


Fig. 8. Analysis for parameter α. (a) Indian Pines image. (b) Salinas image. (c) Pavia Center image.

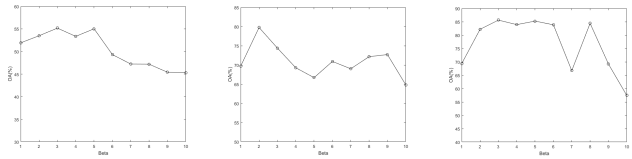


Fig. 9. Analysis for parameter β. (a) Indian Pines image. (b) Salinas image. (c) Pavia Center image.

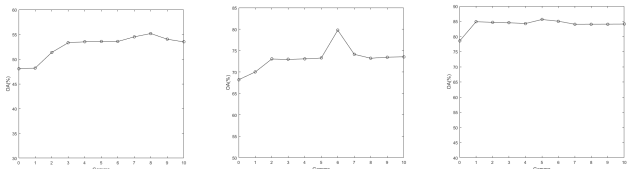


Fig. 10. Analysis for parameter γ. (a) Indian Pines image. (b) Salinas image. (c) Pavia Center image.

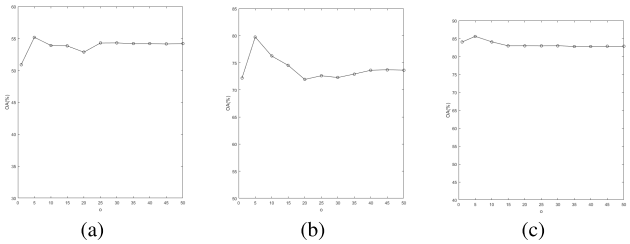


Fig. 11. Analysis for parameter o. (a) Indian Pines image. (b) Salinas image. (c) Pavia Center image.

D. Computational Time

The computational times of the different clustering methods are introduced in this section. The experimental environment of the different clustering methods is as follows: i7-8700K CPU at 3.70 GHz, 3.70 GHz, 32.00 GB RAM, and MATLAB R2016b.

The average computational times of the different clustering methods are shown in Table VII with respect to the three HSIs. Notably, the USPEC and SSSC require less time than the other clustering methods, and SGCNR, RSC, S^5C, BGP-JSDL, and BGP-DL are slower than k-means, FCM,

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TABLE VII
EXECUTION TIME OF DIFFERENT ALGORITHMS (SECONDS)

Data sets	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSSC	S^5C	BGP-JSDL	BGP-DL	BGP-GNMF
Indian Pines	8.51	127.82	163.32	25.81	2.23	62.32	2411.37	5.24	307.01	1665.31	1029.67	196.80
Salinas	23.87	990.82	967.36	-	4.07	91.69	-	40.46	1343.40	4245.99	13938.04	1086.86
Pavia Center	73.14	79.03	1684.03	-	15.10	707.66	-	122.91	5298.25	10526.99	285331.92	8316.08

and FCM_S1, which can be clearly observed from Table VII. The proposed BGP-GNMF is slower than k-means, FCM, FCM_S1, Ncut, USPEC, SSSC, and SGCNR; on the contrary, the clustering accuracy has been improved greatly. The proposed BGP-GNMF is faster than BGP-JSDL, BGP-DL, and RSC. For the Indian Pines image and Salinas image, the proposed BGP-GNMF is faster than S^5C, but for Pavia Center image, BGP-GNMF is slower than S^5C.

V. CONCLUSION

In this article, we propose a novel BGP-GNMF algorithm for coclustering of large HSIs. The proposed algorithm first learns the structured dictionary atoms under the adaptive superpixels and joint sparsity unified optimizing model to explore the adaptive spectral and spatial discriminative features. Then, the orthonormal constrained NMF-based bipartite graph partition is proposed to overcome the challenge in the computational consumption and memory space for the large HSIs in which the learned dictionary atoms and the hyperspectral pixels were simultaneously grouped by the nonnegative indicator matrix. Finally, we incorporate the graph structure into the bipartite graph partition to exploit the intrinsic geometry of the HSIs and further improve the clustering accuracy. Experimental results demonstrated that the BGP-GNMF is efficient and effective in dealing with large HSIs. Although the proposed BGP-GNMF method can achieve good clustering performance, there is still limitation that can be further improved. For example, the proposed BGP-GNMF produces a subspace affinity matrix between hyperspectral pixels and superpixel-based dictionary atoms at first and followed by the spectral clustering for the affinity matrix. In the future, we will incorporate affinity learning and spectral clustering into a unified framework to obtain a global optimal clustering performance.

APPENDIX

Theorem 1: The iterative algorithm in (18) and (19) for the objective function in (15) converges.

Proof: We use the auxiliary function, and assume that $Z(\mathbf{G}, \mathbf{G}')$ is the auxiliary function of $J(\mathbf{G})$. $J(\mathbf{G})$ is described as follows:

$$J(\mathbf{G}) = -2\mathbf{Tr}\left(\mathbf{X}\mathbf{G}\mathbf{F}^T\right) + \mathbf{Tr}\left(\mathbf{F}^T\mathbf{F}\mathbf{G}^T\mathbf{G}\right) + \gamma\,\mathbf{Tr}\left(\mathbf{G}^T\left(\mathbf{I} - \mathbf{U}\Lambda^{-1}\mathbf{U}^T\right)\mathbf{G}\right) + \mathbf{Tr}\left(\Lambda_2\left(\mathbf{G}^T\mathbf{G} - \mathbf{I}\right)\right). \tag{20}$$

It should satisfy the following condition:

$$Z\left(\mathbf{G}, \mathbf{G}'\right) \geq J(\mathbf{G}), \quad Z(\mathbf{G}, \mathbf{G}) = J(\mathbf{G}). \tag{21}$$

To prove that the algorithm converges, we only need to prove $J(\mathbf{G}^{(t)}) = Z(\mathbf{G}^{(t)}, \mathbf{G}^{(t)}) \geq Z(\mathbf{G}^{(t+1)}, \mathbf{G}^{(t)}) \gg J(\mathbf{G}^{(t+1)})$. Thus, $J(\mathbf{G}^{(t)})$ is monotonically decreasing. The key point is to find the appropriate $Z(\mathbf{G}, \mathbf{G}')$ and its global minimum.

The auxiliary function $Z(\mathbf{G}, \mathbf{G}')$ can be described as follows:

$$Z\left(\mathbf{G}, \mathbf{G}'\right) = -2\sum_{ik}\mathbf{G}'_{ik}\left(1 + \log\frac{\mathbf{G}_{ik}}{\mathbf{G}'_{ik}}\right)\left(\mathbf{X}^T\mathbf{F}\right)_{ik} + \sum_{ik}\frac{\left[\mathbf{G}'\left(\mathbf{F}^T\mathbf{F} + \Lambda_2 + \gamma\,\mathbf{I}\right)\right]_{ik}\mathbf{G}_{ik}^2}{\mathbf{G}'_{ik}} + \gamma\sum_{ik}\mathbf{G}'_{ik}\left(\mathbf{U}\Lambda^{-1}\mathbf{U}^T\right)_{ik}\left(1 + \log\frac{\mathbf{G}_{ik}\mathbf{G}_{jk}}{\mathbf{G}'_{ik}\mathbf{G}'_{ik}}\right). \tag{22}$$

The gradient and the Hessian about (22) are described as follows:

$$\frac{\partial Z}{\partial \mathbf{G}_{ik}} = -2\frac{\mathbf{G}'_{ik}}{\mathbf{G}_{ik}}\left(\mathbf{X}^T\mathbf{F}\right)_{ik} + 2\frac{\left[\mathbf{G}'\left(\mathbf{F}^T\mathbf{F} + \Lambda_2 + \gamma\,\mathbf{I}\right)\right]_{ik}\mathbf{G}_{ik}}{\mathbf{G}'_{ik}} - 2\gamma\,\frac{\mathbf{G}'_{ik}\left(\mathbf{U}\Lambda^{-1}\mathbf{U}^T\mathbf{G}'\right)_{ik}}{\mathbf{G}_{ik}} \tag{23}$$

$$\frac{\partial Z}{\partial \mathbf{G}_{ik}\partial \mathbf{G}_{jk'}} = \left[\begin{array}{l} 2\dfrac{\mathbf{G}'_{ik}}{\mathbf{G}_{ik}^2}\left(\mathbf{X}^T\mathbf{F}\right)_{ik} \\ +2\dfrac{\left[\mathbf{G}'\left(\mathbf{F}^T\mathbf{F} + \Lambda_2 + \gamma\,\mathbf{I}\right)\right]_{ik}}{\mathbf{G}'_{ik}} \\ +2\gamma\,\dfrac{\mathbf{G}'_{ik}\left(\mathbf{U}\Lambda^{-1}\mathbf{U}^T\mathbf{G}'\right)_{ik}}{\mathbf{G}_{ik}^2} \end{array}\right]\delta_{ij}\delta_{kk'}. \tag{24}$$

The Hessian matrix in (24) is semipositive, and the global minima are obtained by $\partial Z/\partial \mathbf{G}_{ik} = 0$; we can obtain

$$\mathbf{G}_{ik}^2 = \mathbf{G}'^2_{ik}\frac{\left(\mathbf{X}^T\mathbf{F} + \gamma\,\mathbf{U}\Lambda^{-1}\mathbf{U}^T\mathbf{G}\right)_{jk}}{\left(\mathbf{G}\mathbf{F}^T\mathbf{F} + \gamma\,\mathbf{G} + \mathbf{G}\Lambda_2\right)_{jk}}. \tag{25}$$

The optima of (25) are obtained from (23) and are the global minima of (20). Thus, using (25) to update $\mathbf{G}$ causes $J(\mathbf{G})$ to decrease. Similarly, using (18) to update $\mathbf{F}$ causes $J(\mathbf{F})$ to decrease.

Theorem 2: At convergence, the solutions for (18) and (19) satisfy the KKT optimality condition, that is, the algorithm correctly converges to local optima.

Proof: The Lagrangian function about $J(\mathbf{G})$ is described in (20), and the gradient can be denoted as follows:

$$\frac{\partial J}{\partial \mathbf{G}} = -2\mathbf{X}^T\mathbf{F} + 2\mathbf{G}\mathbf{F}^T\mathbf{F} + 2\gamma\left(\mathbf{I} - \mathbf{U}\Lambda^{-1}\mathbf{U}^T\right)\mathbf{G} + 2\mathbf{G}\Lambda_2. \tag{26}$$

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The nonnegativity constraint $\mathbf{G}$ is enforced using the KKT complementary slackness condition

$$\left(\begin{array}{c}-2\mathbf{X}^T\mathbf{F}+2\mathbf{G}\mathbf{F}^T\mathbf{F}\\+2\gamma\left(\mathbf{I}-\mathbf{U}\Lambda^{-1}\mathbf{U}^{\mathbf{T}}\right)\mathbf{G}+2\mathbf{G}\Lambda_2\end{array}\right)_{ij}\mathbf{G}_{ij}=0. \quad (27)$$

In Theorem 1, we prove the convergence, and the solution satisfies

$$\left(\begin{array}{c}-2\mathbf{X}^T\mathbf{F}+2\mathbf{G}\mathbf{F}^T\mathbf{F}\\+2\gamma\left(\mathbf{I}-\mathbf{U}\Lambda^{-1}\mathbf{U}^{T}\right)\mathbf{G}+2\mathbf{G}\Lambda_2\end{array}\right)_{ij}\mathbf{G}^2{}_{ij}=0. \quad (28)$$

From (27) and (28), we can clearly observe that either the first factor is zero or $\mathbf{G}_{ik}$ is zero. If the first factor is zero, (27) is equal to (28). If $\mathbf{G}_{ik}$ is zero, then $\mathbf{G}_{ik}{}^2$ is also zero and vice versa. Thus, at convergence, the solutions in (18) and (19) satisfy the KKT optimality condition.

From the KTT condition, summing over j, we can obtain $(\mathbf{F}^T\mathbf{F}\mathbf{G}^T\mathbf{G}-\mathbf{F}^T\mathbf{X}\mathbf{G})_{ii}=(\mathbf{F}^T\mathbf{F}\Lambda_1)_{ii}=(\Lambda_1)_{ii}$. For the off-diagonal element of Λ_1,we ignore the nonnegativity constraint and obtain $(\Lambda_1)_{ii'}=(\mathbf{F}^T\mathbf{F}\mathbf{G}^T\mathbf{G}-\mathbf{F}^T\mathbf{X}\mathbf{G})_{ii'}$. Combining the two together, we can obtain $\Lambda_1=\mathbf{G}^T\mathbf{G}-\mathbf{F}^T\mathbf{X}\mathbf{G}$. For Λ_2, similarly, we can obtain $\Lambda_2=\mathbf{G}^T\mathbf{X}^T\mathbf{F}-\mathbf{F}^T\mathbf{F}-\gamma\,\mathbf{G}^T(\mathbf{I}-\mathbf{U}\Lambda^{-1}\mathbf{U}^T)\mathbf{G}$.

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